

Measurement and model limits of allele-specific single-cell perturbation prediction

Abstract

Protein-coding variants of the same gene can produce distinct cellular effects, yet single-cell perturbation prediction is largely formulated at the level of genes, drugs or conditions. It remains unclear whether current measurements and models support prediction at allele resolution. Here we systematically evaluated held-out allele-specific responses across 470 variants in TP53, KRAS, GATA1 and JAK1, comprising 321,043 cells. Conventional direction-based metrics indicated successful response recovery, but allele discrimination remained near chance because predictions predominantly captured gene-shared programs rather than allele-specific residuals. Single-cell reproducibility and replicate-ceiling analyses separated measurement-limited datasets, in which sibling alleles were not reproducibly identifiable, from a model-limited dataset containing resolvable allele-specific structure that current predictors failed to recover. Pilot measurements of 25 to 50 cells per perturbation predicted detection-level rankability on disjoint higher-depth cells. These results establish allele-specific single-cell perturbation prediction as a distinct problem and define when it is measurable, benchmarkable and computationally unresolved.

Protein-coding variants within the same gene can differ in protein folding, biochemical activity, interaction partners and downstream regulation, producing allele-specific cellular consequences^{1,2}. Yet single-cell perturbation prediction has largely been formulated at the level of genes, drugs or experimental conditions³⁻⁵. A growing family of generative, graph-based, optimal-transport and foundation-model approaches predicts responses to unseen interventions⁶⁻¹⁷, and community resources increasingly standardize their evaluation¹⁸⁻²⁰. In these settings, all variants of a gene are typically pooled, represented by the same perturbation identity or omitted entirely. Consequently, a model may recover the average response associated with perturbing TP53 or KRAS without distinguishing which coding variant produced it. Whether current single-cell measurements and models can support prediction at allele resolution therefore remains unresolved.

Allele-specific response prediction is not simply gene-level prediction at finer granularity. Multiplexed assays of variant effect and deep mutational scanning characterize large numbers of coding variants, but generally summarize fitness, activity or molecular function as scalar outcomes^{21,22}. Variant-resolved single-cell assays instead expose population-level transcriptional responses, and continuous protein or biophysical representations make prediction of previously unseen variants technically possible²³⁻²⁵. However, the criteria for success at this resolution are unclear. Correlation or mean squared error can reward a transcriptional program shared by variants of a gene even when the predicted profile carries little information about the causal allele. Detecting that a variant differs from wild type is also insufficient, because several variants may each be detectable relative to a control while remaining indistinguishable from one another. Allele-level prediction therefore requires separating recovery of a shared perturbation direction from identification of the specific allele that generated the response.

This distinction exposes a second unresolved question: when allele discrimination fails, is the limitation experimental or computational? Perturbation benchmarks commonly treat measured response profiles as fixed ground truth and assign uncertainty primarily to the model, although model rankings

can depend on simple baselines, systematic variation and metric geometry^{26,27}. At allele resolution, within-variant sampling variability may approach or exceed the separation between sibling variants. The resulting ground truth may detect perturbation effects without reproducibly encoding allele identity, leaving too little information to reward an allele-specific prediction. Conversely, sufficiently resolved measurements may contain reproducible allele structure that current models fail to infer from molecular features. Distinguishing these measurement-limited and model-limited regimes is necessary both to interpret negative prediction results and to determine which allele-resolved experiments can support meaningful model comparison.

Here we establish held-out allele-specific single-cell perturbation prediction as a distinct evaluation problem and systematically examine its current experimental and computational limits. We integrate 470 protein-coding variants across TP53, KRAS, GATA1 and JAK1, comprising 321,043 cells generated by Perturb-seq, base editing and scSNV-seq. Using AllelePerturb, we separate transcriptional direction recovery from allele discrimination and differential-expression fidelity, and assess whether each dataset supports variant detection, sibling-allele identification and prediction from molecular features. Across feature-based predictors and published variant-conditionable models, direction-based performance was dominated by gene-shared programs, whereas allele discrimination remained near chance. Single-cell reproducibility and replicate-ceiling analyses separated datasets whose measurements did not reproducibly identify sibling alleles from a dataset containing resolvable allele-specific structure that current predictors did not recover. Finally, pilot measurements of 25 to 50 cells per perturbation predicted detection-level rankability on disjoint higher-depth cells. Together, these analyses define the present solvability landscape of allele-specific single-cell perturbation prediction and provide principles for deciding when the task is measurable, benchmarkable and computationally unresolved.

Results

Held-out allele prediction requires joint evaluation of response accuracy and measurement resolution

Held-out allele prediction asks whether the cellular response to an unseen protein-coding variant can be inferred from its molecular properties rather than from its identity or its observed cells. Gene-level formulations cannot pose that question: they assign every variant of a gene the same perturbation identity, so variant responses collapse into a common direction. Allele resolution instead keeps each variant’s departure from wild type intact, so that in TP53 mutations such as R175H, R273C and R248W remain separate prediction targets rather than one pooled response. To formalize this problem, AllelePerturb fixes the prediction target, the admissible input and the held-out design (Fig. 1a). Each perturbation is a protein-coding variant measured at single-cell resolution, and the target is its population-average response, the pseudobulk difference δ_v between the variant’s cells and wild-type cells. A predictor receives only the variant’s molecular features, a six-dimensional biophysical vector θ or a protein-language-model embedding, and never the variant identity or its cells; from these it must produce $\hat{\delta}_v$ for an allele it has not seen. These restrictions define the task rather than an implementation: a method that may read a variant’s identity or its cells is answering a different question.

Deciding whether a prediction succeeds at this resolution cannot rest on a single accuracy number, because allele-level prediction nests requirements that gene-level evaluation never has to separate. A perturbation must be detectable, its cells separable from wild-type cells at all; sibling alleles of the same gene must then be identifiable from one another, since several variants can each be detectable against a control while remaining mutually indistinguishable; and only where identification holds is prediction from molecular features a well-posed question. Model scoring accordingly splits into two distinct axes. Direction recovery (Pearson- δ) asks whether $\hat{\delta}_v$ points the right way in expression space, whereas allele discrimination (PDS) asks whether $\hat{\delta}_v$ is closest to its own variant among the sibling variants of the gene. Neither axis reveals whether the measurement it is scored against encodes allele identity, so detection and identification are assessed directly on the single-cell distributions rather than on the mean. Response accuracy and measurement resolution are therefore read together: a model score becomes interpretable only alongside a verdict on whether the ground truth could reward an allele-specific prediction, that is, on whether a perturbation is measurable, benchmarkable and worth modeling.

Testing these requirements empirically requires variant-resolved data that differ in protein, cell type

and measurement technology. We assembled four such datasets (AllelePerturb-Bench; Fig. 1b,c). TP53 (393 aa, 98 variants) and KRAS (189 aa, 92 variants) were profiled by Perturb-seq in A549 cells²³, GATA1 (413 aa, 254 variants) by base editing in hematopoietic stem and progenitor cells²⁴, and JAK1 (1,154 aa, 26 variants) by scSNV-seq in HT-29 cells²⁵. Together they comprise 470 protein-coding variants and 321,043 single cells, including the dataset wild-type and control cells. Per-variant cell depth varies substantially across genes, with median depths of 929 and 1,000 cells per variant for TP53 and KRAS, 354 for GATA1 and 104 for JAK1 (Fig. 1c), so the available data span a wide range of sampling regimes, which is what makes measurement resolution an empirical question here.

Making variants comparable across genes required representing each one by its protein-level properties rather than by a variant identifier. We encoded variants using a six-dimensional biophysical vector θ , including changes in hydrophobicity, side-chain volume and charge, together with fold-core location, functional-switch residue status and hotspot/pathogenic annotation (Fig. 1d). Because these features describe variant properties beyond gene identity, they place variants from different genes in a shared feature space, so a held-out allele can be predicted from its features alone.

Prediction quality is accordingly separated into three dimensions (AllelePerturb-Eval): direction recovery (Pearson- δ), allele-level discrimination (PDS) and differential-expression fidelity (Fig. 1e). Of these, PDS carries the defining allele-level requirement, whether a predicted response identifies its target variant among other held-out variants; the remaining dimensions can be satisfied by a response that is right about the perturbed gene and uninformative about the allele. Six split strategies then stress-test generalization along complementary axes: random generalization, positional extrapolation, mechanistic extrapolation, cross-gene transfer, low-depth evaluation and compatibility with existing protocols (Fig. 1f). With the task, its data and its evaluation fixed in this way, the question that organizes what follows is whether predictors distinguish alleles at all, or only recover the shared direction of a perturbed gene.

Conventional direction metrics do not establish allele discrimination

Recovery of a perturbation’s direction and identification of the allele that produced it behaved as separable objectives rather than as alternative views of the same success criterion. We evaluated 20 predictors or references, comprising 18 feature-model combinations (six model heads by three feature spaces), the Gene-mean predictor and the WT-null reference (Fig. 2). Allele discrimination remained at chance throughout: across the 19 non-null predictors, PDS-cosine clustered between 0.49 and 0.52 (mean 0.50), and no predictor’s 95% bootstrap confidence interval excluded the 0.50 chance level (Fig. 2b). Point estimates were averaged over 15 independent pseudobulk subsamples to remove single-draw variability, and the interval of every predictor overlapped chance. This verdict did not rest on the analytic 0.50 baseline: within-split label permutation gave an empirical null mean of 0.500 (range 0.495 to 0.504 across 340 method-split-gene combinations), and only 3.2% of combinations exceeded the permutation 95th percentile, consistent with chance under the empirical permutation null. PDS-L1 and PDS-L2 gave the same conclusion.

The same predictors nonetheless recovered the transcriptional direction of perturbation effects. Pearson- δ ranged from 0.55 to 0.65 across predictors, and every non-null predictor had a 95% confidence interval entirely above zero (Fig. 2c). Plotting PDS against Pearson- δ placed every predictor in the same regime, direction recovered and allele ranking failed (Fig. 2d), with method-to-method variation small beside the separation between the two evaluation axes. Success under a conventional direction metric therefore does not establish prediction at allele resolution: variant-level evaluation exposes a systematic direction-ranking dissociation rather than a deficiency of a particular model class.

What the predictors recovered was almost entirely gene-shared. Decomposing each variant’s effect into a gene-shared component and an allele-specific residual, $\delta_v = \bar{\delta}_g + r_v$, left the residual Pearson- δ near zero for every predictor (between -0.08 and 0.03), and a featureless predictor that outputs only the training gene mean already matched the direction score of the best feature model. Removing the shared component therefore removed the recovered signal: the predictions carry the program-level response of a perturbed gene, not the allele-specific residual that separates its variants. Gene-shared program recovery is thus the mechanism behind the dissociation, and in the low-effect datasets it is compounded by residuals that are not reproducibly recoverable: an oracle scored on residual profiles remained at chance for TP53 and KRAS (0.48) and only marginal for GATA1 (0.59), rising only for JAK1 (0.89), whose



Figure 1: The allele-specific prediction problem and its resolution requirements. **a**, Each perturbation is a protein-coding variant measured in single cells, and the target is its pseudobulk response δ_v (variant cells minus wild-type cells). A model sees only the variant's molecular features (θ or a protein-language-model embedding), never its identity or cells, and predicts $\hat{\delta}_v$. Direction recovery and allele discrimination are distinct requirements (**e**). In parallel, a single-cell diagnostic (detection versus wild type, identification versus siblings, on the cell distributions rather than the mean) asks whether the data resolve that distinction at all. **b**, Benchmark coverage: variant positions along the four proteins with domains and hotspot residues; 470 protein-coding variants across three technologies (Perturb-seq, base editing, scSNV-seq). **c**, Per-variant cell depth spans distinct sampling regimes (bars, median cells per variant: 929, 1,000, 354 and 104; 98, 92, 254 and 26 variants and 83.4k, 83.6k, 149.2k and 4.9k cells, including wild-type/control, for TP53, KRAS, GATA1 and JAK1). **d**, Each variant is a six-dimensional biophysical vector θ (changes in hydrophobicity, side-chain volume and charge; fold-core location, functional-switch status, hotspot/pathogenic annotation), placing variants of different genes in a shared space (right, principal components of θ over the 470 variants). **e**, Prediction quality is separated into direction recovery (Pearson- δ), allele-level discrimination (PDS) and differential-expression fidelity. **f**, Six generalization splits test random, positional-extrapolation, mechanistic-extrapolation, cross-gene, low-depth and compatibility settings; bars mark which slice each split holds out, not its exact size, and the cross-gene split holds out two genes, GATA1 and JAK1 (280 of 470 variants). Together these panels establish the task and its nested resolution levels.

variants carry a resolvable allele-specific signal (Supplementary Note).

Differential-expression fidelity gave the dissociation a concrete biological reading (Fig. 2e). Direction agreement reached 65%, whereas DE-LFC Spearman was 0.25 and DE overlap only 15%. Read in that order, the values locate where the signal is lost: 65% agreement on the sign of change, $\rho = 0.25$ for the ranking of effect sizes and 15% for the identity of the responding genes. Predictors therefore reproduce a program-level perturbation direction while losing the finer structure that separates one variant's response from another's.

The size of the dissociation varied across datasets. Defined per gene as the mean Pearson- δ minus the mean PDS pooled over splits (Fig. 2f), it was strongest for TP53 (0.31), intermediate for KRAS (0.19) and GATA1 (0.13), and near zero for JAK1 (-0.04). This ordering anticipated the split-half signal-to-noise analysis below: JAK1 combined the strongest per-variant signal with the weakest dissociation, its two axes nearly coinciding because its per-gene direction recovery is itself lower, consistent with a less gene-shared, more allele-individuated response, whereas TP53 and KRAS lay near the noise floor. Because each gene here is also a distinct dataset, assay and cell type, the gradient is a per-dataset ordering rather than an intrinsic property of the genes (see Discussion).

The same separation reappeared at single-variant scale within TP53 (Fig. 2g). Ridge-esm produced stable Pearson- δ values across representative held-out variants (0.74 to 0.82), whereas PDS for those same variants ranged from 0.00 to 0.95, with more than half of them below chance. A predictor can thus be accurate about the direction of a variant's response while carrying almost no information about which variant produced it. Model choice alone cannot explain this failure, so the remaining explanations must be separated: the design of the evaluation, including the split, the distance metric and standardization, the variant representation and the model interface, and the resolution of the measurement itself. We turn to the evaluation first.

Allele-ranking failure persists across metrics, representations and model interfaces

A chance-level ranking floor could in principle be manufactured by the evaluation rather than encountered in the measurement. If evaluation design set the floor, varying the generalization split should lift PDS; it did not. Across five of the six generalization splits (the sixth, cross-gene, is defined but excluded from the main grid, Methods, because the datasets do not share a gene space), the two evaluation axes moved independently of each other (Fig. 3a,b). Pearson- δ stayed positive under the random, positional-extrapolation and compatibility splits, typically between 0.55 and 0.75, and fell mainly under mechanistic extrapolation and low-depth evaluation. PDS, in contrast, remained inside the empirical chance regime under every split (Fig. 3a,c), degrading most under low-depth evaluation. No split produced a consistent above-chance allele-discrimination regime, so the direction-ranking dissociation is not a property of the random holdout setting.

Neither the geometry of the distance function nor the unit of aggregation moved the chance-level verdict. Cosine, L1 and L2 distances gave nearly identical PDS distributions across the 18 feature-model combinations, all clustered at the chance level of 0.50 (Fig. 3d), so the floor is not an artifact of one metric's geometry. Resolving the same grid by gene and split rather than by method reproduced the pattern (Fig. 3f): Pearson- δ exceeded PDS in 15 of 17 combinations. The two exceptions both arose in small held-out mechanistic-extrapolation sets: the KRAS set, where the metrics nearly coincided (0.71 versus 0.72), and the JAK1 set, where direction recovery itself collapsed (Pearson- $\delta = 0.02$). The lowest value in the grid, a PDS of 0.19 in the GATA1 low-depth split, already anticipates the depth-dependent detection limit quantified below.

Preprocessing choices were equally unable to lift the floor. Refitting the per-gene standardization on training variants and wild-type cells only, excluding every held-out variant, left all predictors at or below chance (no method above 0.50; mean Δ PDS = -0.01 against the reported all-cell standardization) with direction recovery unchanged, so the transductive standardization used throughout is, if anything, conservative with respect to the null (Supplementary Note).

Input representation was the next candidate explanation, and richer variant encodings did not close the gap. Under identical splits, heads and scoring, the six-dimensional biophysical θ vector was compared with 1,280-dimensional ESM-1v embeddings, modern ESM2 embeddings²⁸, three mutation-aware ESM2 constructions that encode the change relative to wild type (the mutant-minus-wild-type global

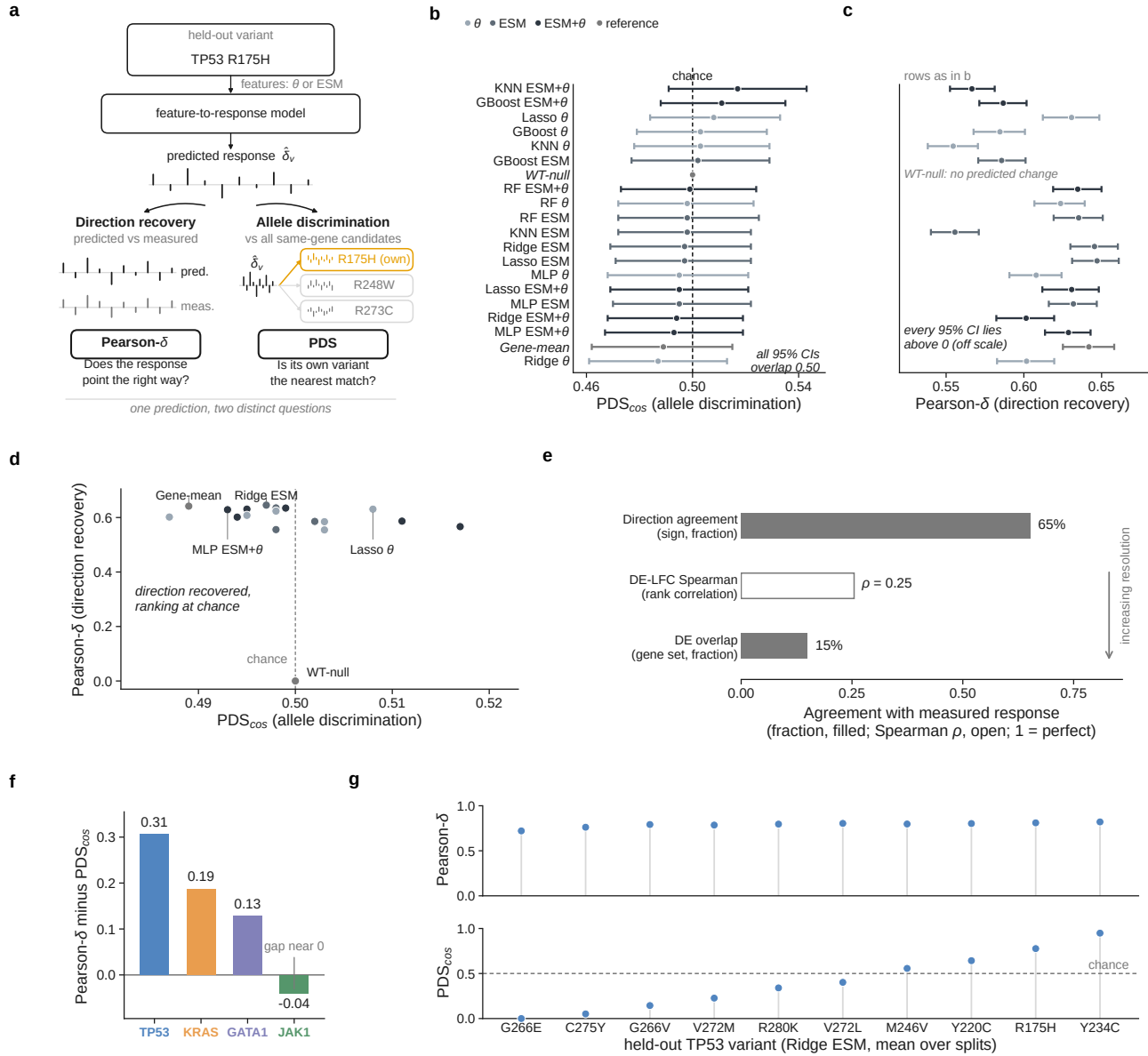


Figure 2: Feature-based predictors recover gene-shared direction without reliable allele discrimination. **a**, How a held-out variant is scored: the predicted response is evaluated for direction recovery (Pearson- δ) and, separately, for allele discrimination (PDS, whether it identifies its own variant among all candidate variants of the same gene, training and held out; Methods). **b**, Allele-level PDS-cosine for 20 predictors; whiskers are 95% bootstrap confidence intervals over held-out variants (Statistical analysis), and every interval overlaps the chance level of 0.50. **c**, The same predictors recover perturbation direction (Pearson- δ), intervals as in **b**; the axis is truncated to 0.52–0.68, so zero is off-scale and every interval lies above it. **d**, Methods cluster in the direction-without-ranking regime. **e**, Differential-expression fidelity: direction agreement 65%, DE-LFC Spearman 0.25, DE overlap 15%. **f**, The dissociation gap (per-gene mean Pearson- δ minus mean PDS-cosine, pooled over splits) follows a per-dataset gradient across the four genes (TP53 0.31, KRAS 0.19, GATA1 0.13, JAK1 –0.04). **g**, Per-variant scores (Ridge- δ , TP53) show stable direction but unstable ranking; each variant is the mean over the splits holding it out (84 evaluations of 59 variants).

embedding, the mutated-residue hidden-state difference and a local-window difference), and AlphaMissense²⁹, a purpose-built structure- and evolution-informed variant-effect predictor. Every representation left allele discrimination at chance (PDS 0.46 to 0.47, single-draw estimates that sit marginally below the 15-subsample values of Fig. 2 but carry the same chance verdict), with similar direction recovery throughout (Pearson- δ 0.63 to 0.66; Fig. 3e, Supplementary Note).

Three controls localize where this representation failure does and does not originate. The direction channel was not feature-driven: a featureless predictor that outputs the training gene mean already reached Pearson- δ 0.66, and shuffling the ESM2 features across variants within a gene left PDS at chance, so the recovered direction is the shared gene-level program rather than allele-specific signal. Nor was the failure a matter of the representation discarding the mutation: the mutated-residue difference was highly variant-specific, with within-gene distances 91% of between-gene distances against 0.1% for the gene-identity-dominated global embedding, yet it too failed. And AlphaMissense pathogenicity did not track the measured transcriptional effect size within any gene (Spearman near zero for TP53, KRAS and GATA1). Mutation-aware and structure-aware representations therefore recover the shared perturbation direction without overcoming the allele-discrimination floor, which does not originate in the input. Whether it originates in the ground-truth measurement or in the mapping from features to responses is a question the input cannot answer, and the analyses that follow resolve it directly (Fig. 4, Fig. 5).

Published perturbation-prediction models occupied the same regime as the feature-regression heads once scored under a matched evaluation. We re-scored external models through a single harness: the same held-out variants, pseudobulk profiles, tie-aware PDS, and the bootstrap and permutation controls used for the in-house heads (Fig. 3g, Supplementary Table 4; Methods). Four of the five variant-conditionable models ran to completion, and each remained at chance: scGen (PDS 0.49, 95% CI 0.46–0.52), scVIDR (0.50, 0.48–0.53), Biolord (0.49, 0.47–0.52) and CellFlow (0.51, 0.48–0.54). None exceeded the feature-model baselines of Fig. 2, whose intervals also overlapped chance, so for the evaluated variant-conditionable models allele discrimination remains unresolved.

The remaining public tools either cannot represent the task or do not clear it. scGPT, GEARS and STATE key their perturbation representation to the identity of the perturbed gene, so these gene-keyed models are allele-blind by construction: all variants of a gene collapse into one prediction, and only 4 of 470 variants (0.85%) are representable at all (Fig. 3h). Consistent with that collapse, the gene-constant Gene-mean reference sits at chance (PDS 0.49). A compositional autoencoder (CPA) fitted with a variant-feature encoder did not converge on the continuous inputs. One generative flow, PerturbNet, marginally exceeded the analytic 0.50 chance level (PDS 0.54), but its predictions are confined to a 50-dimensional expression subspace in which random vectors already score 0.52; its exceedance over a permutation null estimated inside that subspace was not significant (permutation $P = 0.13$), and its direction recovery was negligible (Pearson- δ 0.13). The subspace controls therefore leave this single marginal point uninformative about allele discrimination.

Across this section, five model classes compatible with continuous variant inputs were scored on the task, feature regression, latent arithmetic, disentanglement, optimal transport and conditional-flow generative models, and none achieved above-chance allele discrimination; gene-keyed foundation models could not be scored at all, an interface incompatibility rather than a performance failure, and the compositional autoencoder did not converge. The chance-level verdict thus persists across every model class that can express an allele, and across changes of split, distance geometry, standardization and input representation, which argues against a failure local to any single architecture. Having excluded evaluation design, input representation and model interface as sufficient explanations, the remaining question is whether the experimental ground truth reproducibly encodes allele identity (Fig. 4).

Split-half and sibling diagnostics identify measurement-limited regimes

With the evaluation itself excluded, one alternative remains: predictors may fail to rank alleles not because the models are wrong but because the ground-truth variant responses are not stably separable at the available depth. Deciding this requires separating the task into three nested levels of resolution, each a precondition for the next (Supplementary Table 3): *detection*, whether a variant is distinguishable from wild type; *identification*, whether a variant is distinguishable from the sibling alleles of the same gene; and *prediction*, whether the response of an unseen allele is recoverable from its molecular features.

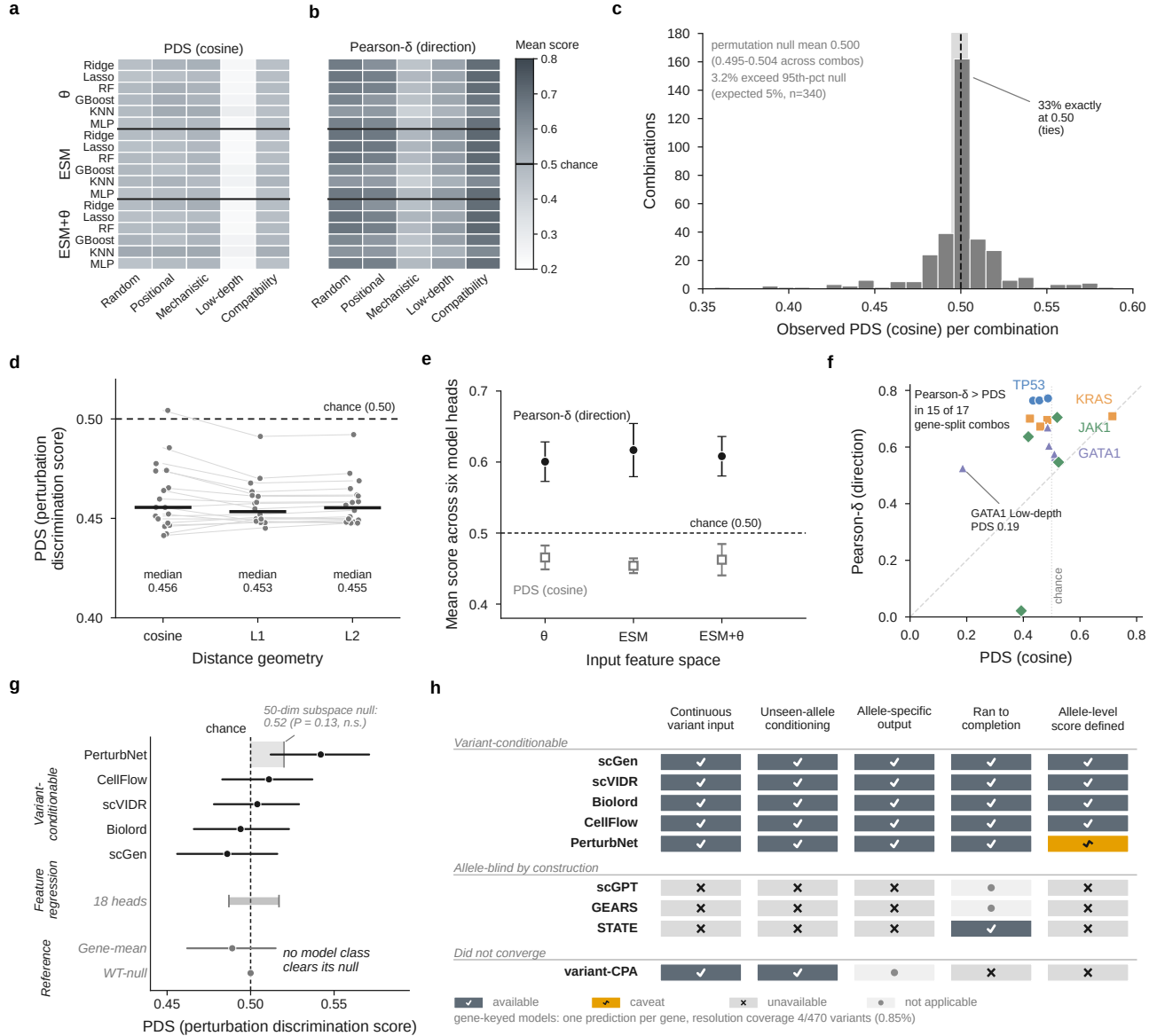


Figure 3: Allele-ranking failure persists across evaluation choices and model interfaces. **a**, PDS across five generalization splits stays near chance, with no split showing a consistent above-chance allele-discrimination regime and the strongest collapse under the low-depth split. **b**, Pearson- δ stays consistently positive across the same splits. **c**, Empirical permutation null: within-split label permutation centers PDS at 0.50 (per-combination null means 0.495–0.504), with observed exceedances of the 95th percentile at 3.2% (consistent with chance), so chance-level PDS is not an artifact of the analytic baseline. **d**, Ranking failure is invariant to the distance function (cosine, L1 and L2 all near chance across the 18 feature-model combinations). **e**, Richer features do not close the ranking gap (θ , ESM and ESM+ θ all near 0.50); error bars are s.d. over the six model heads. **f**, The dissociation holds across all gene-by-split combinations, with Pearson- δ exceeding PDS in 15 of 17. **g**, Published perturbation models scored under one identical harness (same held-out variants, pseudobulk profiles, tie-aware PDS and bootstrap/permutation controls): every variant-conditionable model except PerturbNet, the in-house feature-regression range and both references overlap the PDS = 0.50 chance line; PerturbNet’s full-harness estimate exceeds 0.50, but random predictions confined to its 50-dimensional output subspace already reach an elevated PDS, and its in-subspace exceedance over the corresponding permutation null is not significant ($P = 0.13$) (Supplementary Table 4). **h**, Interface-compatibility audit: gene-keyed models (scGPT, GEARS, STATE) are allele-blind by construction, resolving only 4 of 470 variants (0.85%); this is interface incompatibility, distinct from performance failure.

Detection is necessary but not sufficient, because several variants can each depart from wild type while remaining mutually indistinguishable. Identification is the level that sets the evaluability floor for allele prediction, because a metric can reward a model only for separations that the ground truth itself resolves. Locating a dataset on these levels turns a negative prediction result into a positioned result: it states where the task currently stops, and whether the next move belongs to the experiment or to the model.

Detection was quantified first, directly on the single-cell distributions rather than on the pseudobulk prediction target (Fig. 4a,b). For each variant, cells were randomly divided into two halves, and the energy distance between the two halves of the same variant (D_{self}) was compared with the distance between that variant and wild type (D_{null}). A $D_{\text{self}}/D_{\text{null}}$ ratio near 1 indicates that within-variant sampling variation is comparable to the variant-to-wild-type signal, so the detection window is closed; a ratio well below 1 indicates an open window, in which the perturbation effect stands above sampling noise.

The four datasets occupied distinct detection regimes (Fig. 4c). TP53 ($D_{\text{self}}/D_{\text{null}} = 0.96$, 95% CI 0.95–0.98), KRAS (1.00, 0.99–1.02) and GATA1 (0.88, 0.85–0.90) sat near the floor, whereas JAK1 was well separated from it (0.21, 0.12–0.33), reproducing the ordering seen in the direction-ranking dissociation of Fig. 2f. This per-dataset gradient describes the measurement regimes available here rather than an intrinsic property of the proteins. For the three low-effect datasets detection was at best marginal: the variant-to-wild-type shift is comparable to within-variant replicate noise ($D_{\text{self}}/D_{\text{null}} \approx 1$), so although a weak split-half criterion counts a slim majority as detectable (Fig. 4g), the signal barely clears the per-variant floor, and clearing it licenses no claim about alleles.

Identification, the stricter level that actually bounds allele benchmarking, then failed. A model-free classifier trained to separate the single cells of two sibling variants under cross-validation was at chance for TP53 and KRAS and near chance for GATA1 (median identification AUROC 0.49, 0.50 and 0.52, against 0.50 under label permutation and a wild-type-versus-wild-type control), rising only for JAK1 (0.87; Fig. 4e). Distributional distances agreed: the transcriptional distance between sibling alleles fell below within-variant replicate noise for almost every pair in TP53, KRAS and GATA1 (median nearest-sibling cosine distance 0.23 to 0.46 versus replicate noise near 0.97), leaving 0%, 0% and 3.8% of allele pairs resolvable, whereas the threefold-lower replicate noise of JAK1 left 78.5% resolvable (Fig. 4d; at the single-nearest-sibling level JAK1 identification is a benchmark-level rather than a per-variant property). The GATA1 measurement therefore sits between these outcomes: detection is marginal, the pairwise signal is weak, and no sibling allele is reliably recovered as a variant's nearest neighbour. Only the JAK1 dataset cleared identification under the available assay and protocol, which makes it the only dataset here whose ground truth can reward an allele-specific prediction at all.

Detection and identification degraded together across these datasets, but they remain distinct levels: detection-level rankability, the split-half signal-above-replicate-noise criterion used for Fig. 4g,h and for the atlas comparison below, is necessary but not sufficient for identification, and a dataset can clear detection while still leaving a benchmark nothing to score. The identification verdict was invariant across five distance families, from energy and maximum-mean-discrepancy to sliced-Wasserstein, cosine and Euclidean, so it is not an artifact of energy-distance geometry (Supplementary Note). It is equally not an artifact of the pseudobulk prediction target: the classifier and these distributional distances operate on the single cells themselves, so the discrimination floor is present in the full response distributions and not only in averaged profiles, and averaging is therefore not what erases allele identity (Supplementary Note). What bounds allele benchmarking is not the nominal number of cells but the effect-size-to-noise window available for identification.

Effect size and depth taken together show why that window, rather than cell number, sets the floor (Fig. 4f,g). JAK1 had the lowest nominal depth among the four (median 104 cells per variant) but the widest effect-size-to-noise window, and it was already saturated in the split-half detection assay at 50 cells per half. TP53 and KRAS were profiled far more deeply yet had narrow allele-specific effect windows, remaining close to the noise floor at median raw depths of 929 and 1,000 cells per variant. In the canonical PCA-50 space, subsampling from 50 to 300 cells per half preserved the same ordering: TP53, KRAS and GATA1 stayed below 75% detectable across the evaluated depths, whereas JAK1 re-

mained fully detectable, although its number of eligible variants decreased at the higher depths. Within the tested range of 50 to 300 cells per half, additional cells improved detection only modestly for TP53, KRAS and GATA1 and left JAK1 already saturated. Depth is therefore not the quantity to optimise when the allele-specific effect is small relative to sampling and biological variability, a statement about the assays and protocols evaluated here rather than a depth requirement that transfers to other measurements.

At native depth, and scored over every perturbation rather than over matched subsamples, the ordering persisted. Under the same native-depth, all-perturbation, direct split-half energy-distance definition, the three low-effect allele-level datasets (TP53, KRAS and GATA1) were 98–100% un-rankable by this detection-level split-half criterion, whereas JAK1 remained a positive control with only 10% un-rankable variants (Fig. 4h). Results were qualitatively robust under alternative rankability criteria, including bootstrap-CI-based, increased-stringency and ratio-threshold definitions (Supplementary Fig. 1). The same detection-level floor was also present in external gene-level Perturb-seq atlases, motivating the prospective design analysis below: finite-depth distributional distances can leave a substantial fraction of individual perturbations below the resolution required to score them, in gene- and drug-level screens as well as in allele data. This per-perturbation floor is distinct from the benchmark-level ranking resolution examined next, to which many individually un-rankable perturbations can still contribute in aggregate (Fig. 5).

Read across the detection, identification and prediction levels, these diagnostics locate each dataset on a solvability map. As measured here, TP53 and KRAS are measurement-limited: allele identity is not reproducibly encoded at the available depth, so a benchmark built on them cannot reward an allele-specific prediction. GATA1 lies at the boundary of the same regime. JAK1 is identifiable under its assay and protocol, which makes it the only dataset here in which an allele-level score reports on the model rather than on the measurement. What remains is to quantify how much allele discrimination the measurement itself permits, and what follows for the benchmark when that quantity sits at the floor.

Replicate ceilings link measurement resolution to model-ranking stability

With the evaluation itself excluded and a ground-truth resolution floor established, two questions follow. How much allele discrimination can the measurement support, and can a benchmark at that resolution recover the true ordering of models? The first question needs a reference point that is measured rather than assumed, so we scored an oracle predictor whose prediction for each held-out variant is an independent second measurement of that same variant, taken from a disjoint split-half pseudobulk and passed through the identical harness (Methods). The resulting quantity is an empirical reproducibility reference, or ceiling, at the available depth and protocol: it states how closely one measurement of an allele agrees with a second measurement of the same allele, which is the most that a benchmark scored against that measurement can reward. For TP53 and KRAS this reference was itself at chance (PDS 0.485, 95% CI 0.445–0.525, and 0.500, 0.440–0.566). Split-half depth was not the explanation: the value was unchanged when all cells were used (0.486 and 0.492), and a near-true prediction, the deep true mean of a variant scored against the benchmark’s own noisy measurement, also remained at chance (0.49 and 0.49; Supplementary Note). In these two datasets the oracle reference, the best feature-regression head (a maximum over 18 models, upward-biased under the null) and the near-true-prediction control are all statistically indistinguishable from 0.50 with overlapping 95% confidence intervals. Above-chance allele discrimination therefore could not be reliably demonstrated at this resolution, and the reason is specific: the measurement cannot reliably reward even an essentially correct answer, so the binding limit lies in the resolution of the ground truth rather than in model capacity.

The reference was not at the floor everywhere, and its spread across datasets separates two regimes that a model score on its own cannot distinguish. GATA1 sat marginally above chance (0.572, 0.549–0.595), consistent with its small resolvable-pair fraction and with the marginal detection window above. In the JAK1 dataset, by contrast, the reference reached 0.792 (0.742–0.838) while the best in-house model stayed near chance (0.52), so the measurement there is adequate for allele discrimination and the short-fall is computational (Fig. 5b,c, Supplementary Table 5). That regime is evaluable but unresolved: none of the predictors evaluated here approached the reproducibility reference under the available assay and protocol. The more favourable JAK1 window is a property of that per-dataset measurement regime rather than evidence of stronger allele biology. The reference has a defined scope of its own. It is an em-

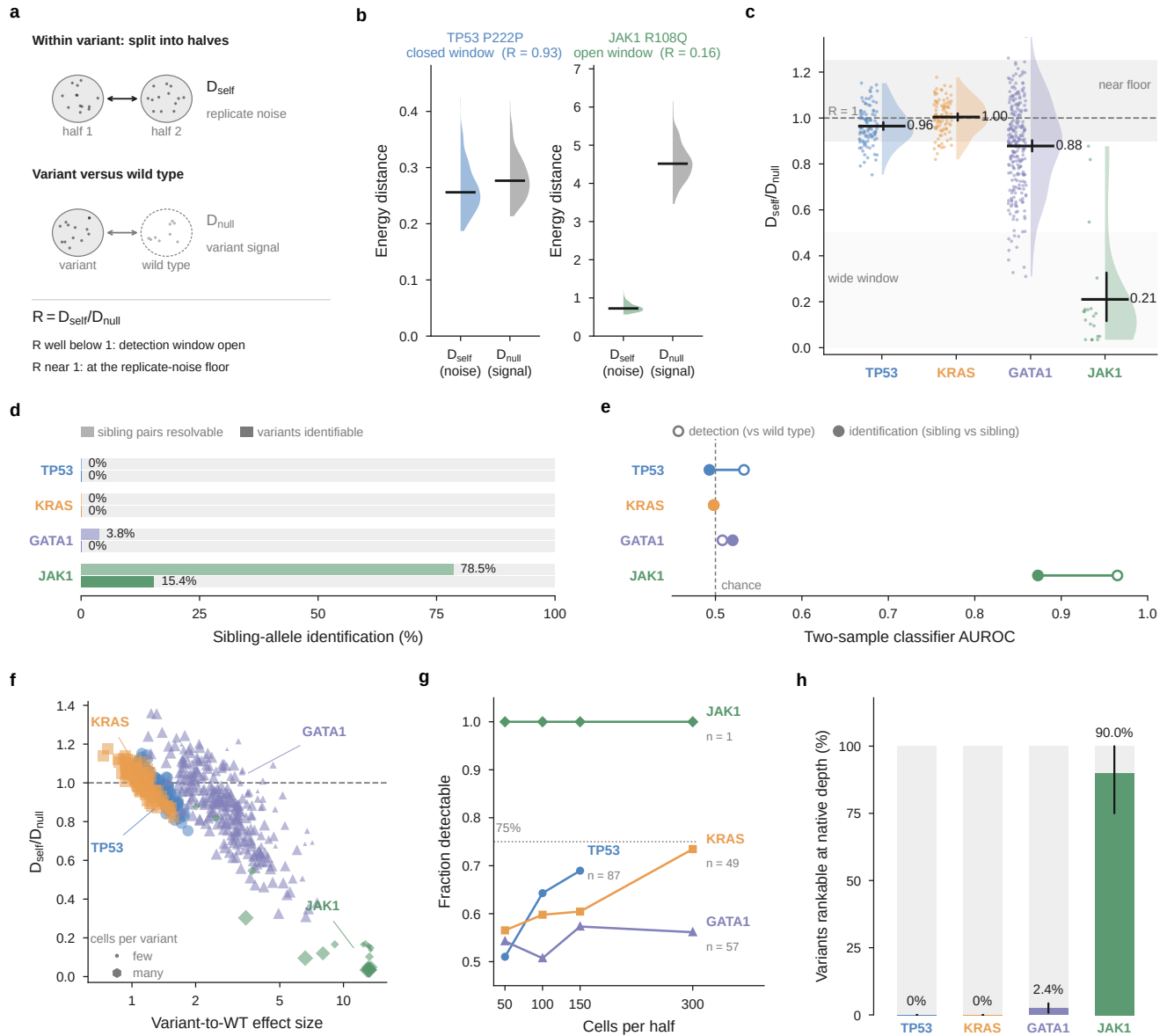


Figure 4: Split-half analysis reveals a marginal detection window and absent allele identification. **a**, Definition of the detection window from within-variant (D_{self}) and variant-to-wild-type (D_{null}) distances. **b**, Representative variants illustrate a closed window (TP53 P222P, $D_{self} \approx D_{null}$) and an open window (JAK1 R108Q, $D_{self} \ll D_{null}$); R here is a 200-seed split-half estimate for one variant, not the native-depth statistic of **c**, **f** and **h**, and the two sub-panels have independent y scales. **c**, Per-variant D_{self}/D_{null} places TP53 (0.96), KRAS (1.00) and GATA1 (0.88) near the detection floor and JAK1 (0.21) well below it; $n = 98, 92, 254$ and 20 variants (six JAK1 variants have 33–49 cells, too few to split), bar and whisker are the gene mean and its 95% bootstrap interval, and the bands mark ratios 0.90–1.25 (replicate-noise floor) and 0–0.50 (window open). **d**, Allele identification: the fraction of sibling-variant pairs whose transcriptional distance exceeds within-variant replicate noise (TP53 and KRAS 0%, GATA1 3.8%, JAK1 78.5%) and the stricter fraction of variants whose single nearest sibling is separable (JAK1 15.4%, 0% for the other genes). **e**, Classifier two-sample test: cross-validated AUROC separating the single cells of sibling variants (identification) is at chance for TP53 and KRAS (0.49, 0.50), near chance for GATA1 (0.52) and high for JAK1 (0.87); detection against wild type (open symbols) is likewise near chance for the low-effect genes and high only for JAK1; label permutation gives 0.50. **f**, Effect size and effective depth jointly determine the window, not cell number alone. **g**, Depth titration in the canonical PCA-50 space: detection rate rises with cells per half in a gene-dependent manner (JAK1 saturated at 50 cells, others below 75%); $n = 14, 9, 6$ and 1 JAK1 variants at 50, 100, 150 and 300 cells per half. **h**, Fraction of variants detection-rankable at native depth, i.e. split-half signal above replicate noise (TP53 0%, KRAS 0%, GATA1 2.4%, JAK1 90%; same n as **c**); whiskers are 95% bootstrap intervals over variants, degenerate at exactly 0 and 100%. Gene-level atlases are compared in Fig. 6f.

pirical reproducibility estimate at the available depth and protocol, not a mathematical upper bound on all denoising or hierarchical models: a model that pooled information across variants could in principle exceed a single replicate, although the benchmark as measured could not reward it.

The second question is different in kind, and the distinction is easy to lose. Whether an individual perturbation is separable from its siblings is a per-perturbation property of that perturbation's effect size and noise; whether a benchmark can order competing models is a property of the perturbation set as a whole, and the two need not agree. To probe the set-level property we built synthetic predictors of known, graded quality by interpolating between the gene-mean profile and each variant's measured effect (Fig. 5d,e), then asked how often the observed PDS recovered their true ranking under bootstrap resampling of the held-out variants (Methods). When the oracle window was at or near chance (ceiling ≤ 0.52 , comprising TP53, all evaluated KRAS depths and the shallowest GATA1 configuration), the true predictor order was recovered only 27% of the time and the best predictor selected only 38% of the time, far from reliable. Once the window opened modestly across the evaluated GATA1 depth configurations (ceiling 0.52–0.55), correct ordering rose to 95% and correct winner selection to 97%; above a ceiling of 0.65 both approached 1.0 (Fig. 5g). These transition points are empirical to the present candidate-set sizes and distance metric rather than fixed thresholds, and recovery also fell when few variants were held out, as expected from test-set size. The consequence is a validity condition rather than a description: below this resolution a leaderboard is unstable whatever the models' true quality, so the ranking it prints reports the measurement as much as the methods (Fig. 5g).

Across benchmarks, this set-level resolution varies widely. Applying the same analysis to five diverse public perturbation benchmarks at a fixed shallow depth of 50 cells per half for comparability, Norman, Replogle, Adamson and the Virtual Cell Challenge training set all recovered the true predictor ordering (resolution 1.00) and a drug screen (sci-Plex³⁰) reached 0.77, whereas TP53 and KRAS were the only datasets at the floor (0.01 and 0.02; Fig. 5f). Set-level ranking resolution and the per-perturbation un-rankable fraction of Fig. 6f are therefore separate quantities, and a dataset can score well on one while failing the other: an atlas such as Replogle can leave roughly half of its individual perturbations un-rankable and still order graded predictors, because many weak perturbations aggregate across thousands of interventions, whereas each allele dataset here contributes at most a few hundred. The same arithmetic works in the allele setting, where the many variants of GATA1 (0.51 at this shallow depth, versus 0.95 at its native ceiling in Fig. 5g) partly offset a low per-variant signal. The position of the allele datasets at the floor thus reflects a lower ground-truth resolution than that of diverse gene- and drug-level screens, rather than a peculiarity of the evaluation. The two questions of this section then converge on one conclusion for the field: allele-specific prediction cannot be judged from model scores alone, because the same score has different meanings in a measurement-limited and in a model-limited regime, and which regime applies is fixed by the measurement before any model is trained. Anticipating that resolution from pilot-scale quantities, before a screen is committed, therefore becomes the practical priority.

Pilot measurements prospectively predict perturbation rankability

If allele discrimination is meaningful only above a measurement floor, then which perturbations are evaluable has to be settled before any model comparison begins. Whether that evaluability can be anticipated from quantities available before modelling, and before a full screen is committed, is the practical question the floor raises. Using leave-one-dataset-out (LODO) cross-validation across the four allele-resolved datasets assembled here and three external Perturb-seq atlases, we trained a rankability predictor whose single feature is each perturbation's pilot-estimable pseudobulk effect size (Fig. 6a; Methods).

A single effect-size feature was sufficient to separate rankable from un-rankable perturbations. Across the five datasets in which rankability is non-degenerate, it did so with a mean AUROC of 0.96 (Replogle 0.97, Norman 0.93, Adamson 0.93, GATA1 0.95, JAK1 1.00; ROC curves and 95% bootstrap intervals in Fig. 6b,c). The TP53 and KRAS datasets could not be evaluated at all: at native depth every one of their variants is un-rankable, and that single-class outcome is itself the clearest available statement, that under the assay, cell context and depth of these experiments the measurements lie entirely below the detection floor. Effect size is therefore a usable triage signal precisely where rankability varies; where it does not vary, the floor has already foreclosed the comparison.

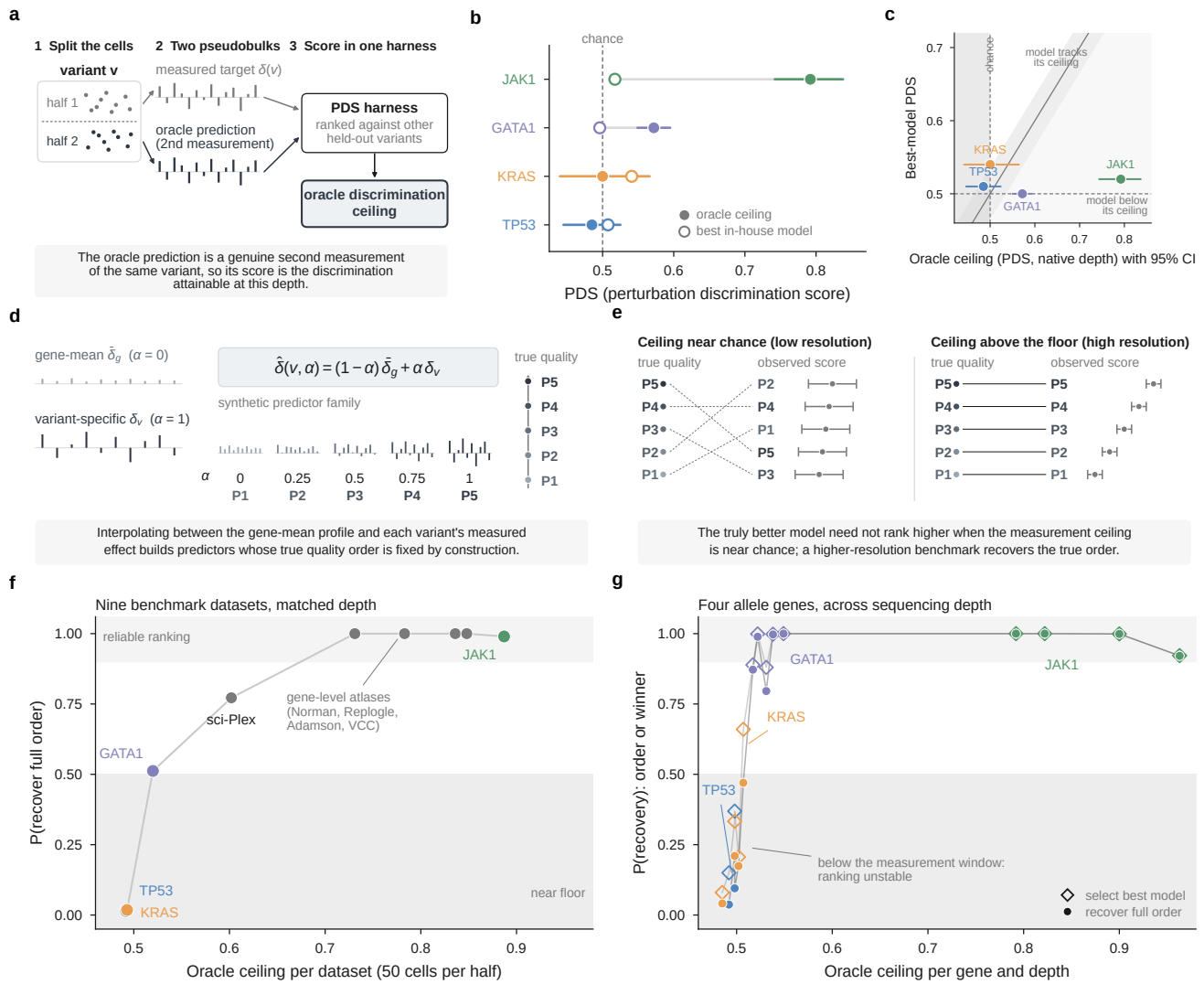


Figure 5: Replicate resolution governs attainable discrimination and benchmark validity. **a**, The oracle predictor defined: an independent second split-half measurement of each held-out variant, scored through the identical harness, estimating the discrimination attainable at the available depth. **b**, Oracle ceiling per gene (native depth) versus the best in-house model; whiskers are 95% bootstrap intervals over held-out variants. TP53 and KRAS ceilings are at chance (measurement-limited); the best model's slight excess over the ceiling reflects best-of-18-heads selection noise, its interval also overlapping 0.50, not a real gap. The JAK1 ceiling is high while models remain near chance (a residual computational gap). **c**, Ceiling-versus-model phase map; bars are 95% bootstrap intervals on the ceiling and chance is the only threshold drawn. A ceiling whose interval covers 0.50 cannot score any model (measurement-limited: TP53, KRAS). The shaded diagonal is the benchmark-solvable band, where a model sits within 0.035 PDS of its ceiling; below it the ceiling clears chance but the model does not follow (GATA1, JAK1). GATA1 clears chance on this oracle-PDS diagnostic while its detection and identification diagnostics stay at the floor (Fig. 4c,d,h). **d**, Construction of synthetic predictors of known, graded quality by interpolating between the gene-mean profile and each variant's measured effect, fixing the true ordering by design. **e**, Illustration of benchmark validity: the same true model order is scrambled by a near-floor benchmark but recovered by a higher-resolution one. **f**, Benchmark resolution across nine perturbation datasets: the probability of recovering the true ranking of graded synthetic predictors (resolution) versus each dataset's oracle ceiling (estimated at a matched 50 cells per half). The four allele genes and five external gene- and drug-level screens trace a phase transition, collapsing at the floor (TP53, KRAS) and approaching one above a ceiling of ~ 0.65 ; bands mark recovery ≥ 0.90 and ≤ 0.50 and are reading aids, not analysis thresholds, here and in **g**. **g**, The benchmark-validity transition: full-order recovery and best-model selection probabilities are both low below the resolution threshold (TP53, KRAS) and jump toward one once the ceiling clears 0.52 (GATA1) and beyond (JAK1); selecting a winner leads recovering the full order.

Because the retrospective predictor draws its feature and its label from the same cells, we next tested it prospectively on cells that the pilot never saw. Pilot measurements of 25 to 50 cells per perturbation were used to predict rankability at higher depth on the disjoint remainder, again under leave-one-dataset-out cross-validation. Across the three benchmarks with balanced labels (Adamson, Norman and Replogle) the pilot predictor reached AUROC 0.85, 0.94 and 0.99 (mean 0.93), with near-diagonal calibration and a label-permutation null of 0.49; a training-free predictor using the pilot signal-to-noise ratio performed comparably (0.84 to 0.98 per dataset), and the result held when the pilot was restricted to 25 cells (Fig. 6d). Detection-level rankability is therefore anticipated from a small pilot, on cells that do not enter the evaluation.

One boundary condition qualifies the triage. What the pilot estimates is perturbation-versus-reference separation, that is detection-level rankability, which is necessary but not sufficient for allele benchmarking: a perturbation can clear the detection floor and still lack the sibling-level separation that identification requires (Fig. 4d,e). Pilot rankability therefore excludes perturbations that are clearly unmeasurable rather than certifying that allele identity is recoverable, and it orders candidate perturbations relative to one another within a given assay rather than fixing a depth that transfers across assays.

Beyond triage, the same effect-size analysis recasts experimental depth as a quantity conditioned on effect size rather than fixed in advance (Fig. 6e). In the JAK1 dataset, whose measured responses are comparatively large under its assay and cell context, detection cleared the floor from as few as 50 cells per half, whereas the TP53 and KRAS measurements remained near the noise floor despite median raw depths of 929 and 1,000 cells per variant. The design target is therefore not a fixed cells-per-variant threshold, but the depth demanded by a particular combination of effect size, inter-variant difference and sampling noise. The binding constraint is the effect-size-to-noise window the experiment must clear, not the raw cell count.

How much the design framing matters is visible in the external gene-level atlases (Fig. 6f). At native depth, 55.3% of perturbations were un-rankable in Replogle 2022⁴ (95% CI 52.8–57.5%), compared with 14.6% in Adamson 2016³¹ and 3.4% in Norman 2019³². Comparing like with like, on the perturbations measured at both depths, normalizing to the same shallow depth ($n = 50$ per half) left every dataset less evaluable and moved the un-rankable fractions toward one another: Replogle rose from 52.0% to 54.9% ($n = 1,299$), Adamson from 14.6% to 38.5% ($n = 96$) and Norman from 2.7% to 11.2% ($n = 224$). Depth normalization can therefore change substantially which perturbations are evaluable, while dataset-specific effect-size structure remains important.

Taken together, these analyses point to design principles rather than a fixed protocol (Fig. 6g). A small pilot, scored on cells that are then set aside, can indicate which perturbations are worth profiling more deeply. Perturbations that fall below the detection floor are better redesigned, measured more deeply, or reported separately, than entered into a model comparison their ground truth cannot arbitrate. Model comparison is informative only on perturbations whose measurements support the distinction being claimed: detection-level rankability licenses questions about perturbation versus reference, whereas allele discrimination additionally requires that sibling alleles be separable in the measurement itself. Evaluation at allele resolution is therefore better organized as a power-aware, stratified design over perturbations than as a uniform ranking of models.

Discussion

Allele-specific single-cell perturbation prediction is not gene-level response prediction at finer resolution. At allele resolution, recovery of a shared perturbation program and identification of the causal variant become separable objectives, and the evaluated predictors met the first while failing the second: they reproduced the transcriptional direction associated with perturbing a gene while carrying little information about which coding variant had produced that response. A high correlation with the measured response therefore does not, by itself, establish allele-specific prediction. Two controls fix its interpretation. A featureless gene-mean baseline, which by construction assigns every variant of a gene the same profile, already reached the direction score of the best feature-based model, so that channel is available without any allele information at all. And when each measured effect was decomposed into

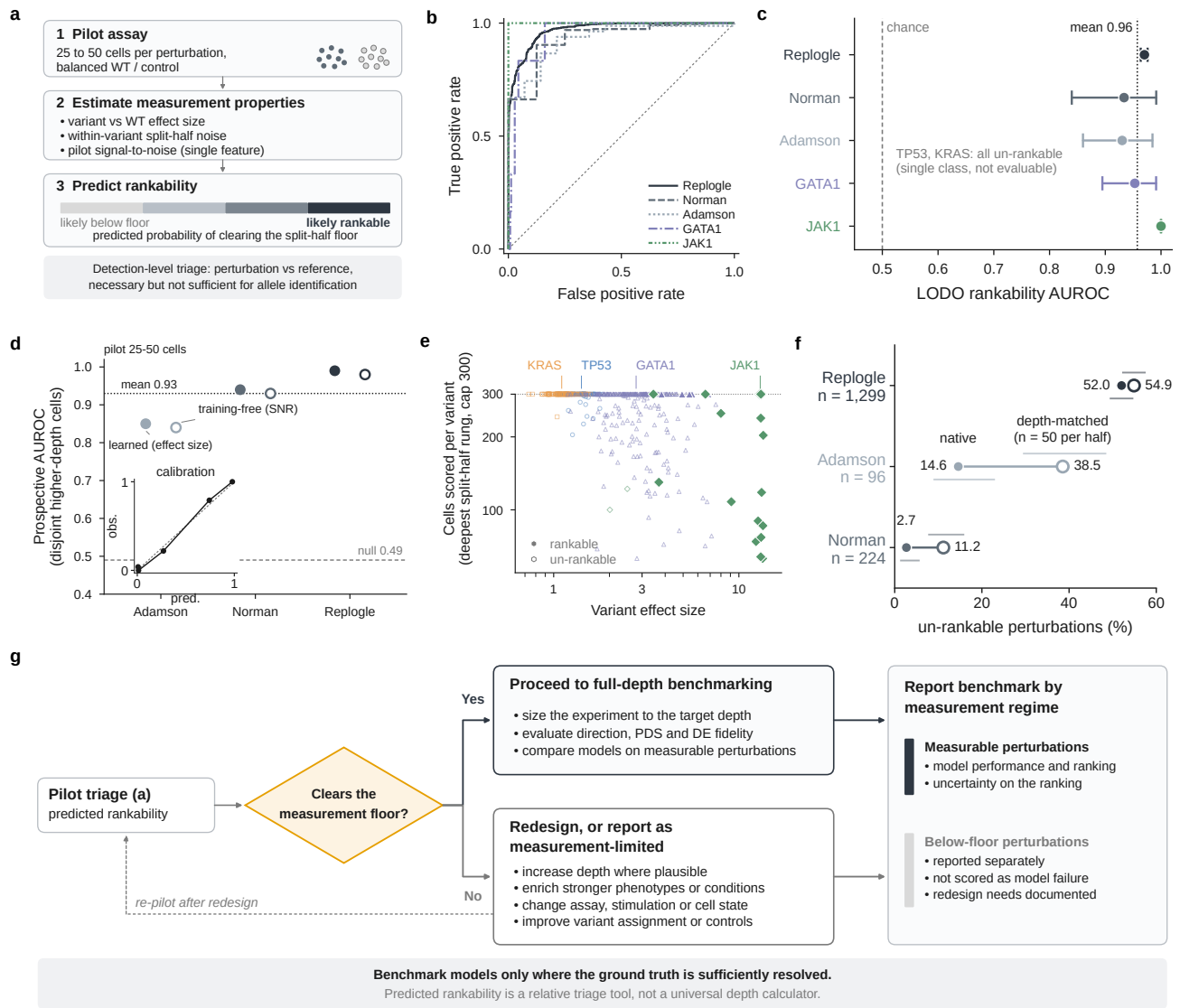


Figure 6: A power-aware reporting protocol for allele-resolution perturbation prediction. **a**, Rankability prediction: each perturbation's pilot-estimable effect size triages which perturbations clear detection-level rankability before any model comparison begins. **b**, Leave-one-dataset-out ROC curves of the single-feature (log effect-size) rankability predictor. **c**, Dataset-level LODO AUROC (mean 0.96): Replogle 0.97, Norman 0.93, Adamson 0.93, GATA1 0.95, JAK1 1.00; whiskers are 95% bootstrap intervals over held-out perturbations; TP53 and KRAS are uniformly un-rankable at native depth (single class), so their AUROC is undefined and they are not evaluable. **d**, Prospective validation: from a 25 to 50-cell pilot, rankability at higher depth is predicted on disjoint held-out cells under leave-one-dataset-out cross-validation (AUROC 0.85, 0.94 and 0.99 for Adamson, Norman and Replogle; label-permutation null 0.49), with near-diagonal calibration. **e**, Effect-size-conditioned design landscape: rankability is set by the effect-size regime, not by a fixed cell-count cutoff (filled, rankable; open, un-rankable). **f**, Native versus depth-matched (n = 50 per half) un-rankable fraction on external gene-level atlases, both arms over the perturbations measured at both depths so they are paired (n = 1,299, 96 and 224); thin bars are Wilson 95% intervals. All three become less evaluable at matched depth (Replogle 52% to 55%, Adamson 15% to 39%, Norman 3% to 11%), Replogle highest throughout, indicating dataset-specific effect-size structure beyond depth. **g**, The design and evaluation principles that follow: estimate effect size and the split-half floor in a pilot, triage perturbations by predicted detection-level rankability, size the experiment to the effect-size-to-noise window it must clear, and compare models only where the ground truth resolves the distinction being scored.

a gene-shared component and an allele-specific residual, direction recovery on the residual was near zero, so removing the shared program removed the recovered signal. What the correlation rewards is the program-level response of a perturbed gene, not the residual that distinguishes its variants. Allele-level prediction must consequently be defined jointly by the mapping from variant features to cellular response and by the resolution of the experimental measurement used to evaluate that mapping.

Ordering the task into three nested questions makes the present solvability landscape legible. Detection asks whether a perturbation differs from its reference condition; identification asks whether related perturbations are reproducibly distinguishable from one another; prediction asks whether the response to an unseen perturbation can be inferred from its molecular features. These questions are ordered, and the ordering has consequences: detection does not guarantee identification, and prediction cannot be meaningfully evaluated where the measurement does not support identification. Read through this hierarchy the four datasets did not fail in the same way, and the distinction they draw is between measurement-limited and model-limited regimes. For TP53 and KRAS, above-chance allele discrimination could not be reliably demonstrated under the available measurement and scoring protocol, and the replicate ceiling was itself indistinguishable from chance, so the binding limit lay in the resolution of the ground truth rather than in model capacity. GATA1 sat just above that floor rather than across it: detection was marginal, the sibling signal was weak, and a variant's single nearest sibling was not reliably identified, so its measurements were not yet adequate to reward an allele-specific model. Only the JAK1 dataset cleared identification under the available measurement regime. In the JAK1 data an independent second measurement of the same variants recovered allele identity that none of the evaluated predictors recovered from molecular features. JAK1 therefore defines an evaluable but unresolved, model-limited regime, not evidence that allele-level prediction is intrinsically impossible. Because each gene here is also a distinct assay, cell type, stimulation condition and analysis protocol, these differences describe dataset-specific measurement regimes and not intrinsic properties of the genes or a ranking of their allele biology. The task is therefore not uniformly out of reach: allele-resolved datasets currently occupy different solvability states, and which state a dataset occupies is an empirical property that can be established before any model is compared.

These distinctions have implications for virtual-cell benchmarking more generally. Existing benchmarks commonly treat experimentally derived response profiles as fixed ground truth and place uncertainty primarily on the model^{3,18–20}, and recent work has shown that rankings can also depend on baseline choice, systematic variation and metric geometry^{26,27}. Measurement resolution is a further and logically prior source of uncertainty: an experimentally derived response profile is an estimate rather than a noiseless label, so the resolution of a task has to be established before models are ranked on it. A benchmark can compare models reliably only when its measurements preserve the distinctions that define the task. Below that resolution, increasing model complexity cannot create evaluable information, and a leaderboard may be unstable even when the predictors it orders differ in underlying quality. Direction-based leaderboards carry a further hazard that is independent of depth: because a shared perturbation program is recoverable without allele information, ranking by correlation can reward that shared program, and can reward a prediction that resolves nothing at the intended granularity. Benchmark design should therefore distinguish two resolutions. Perturbation-level resolution determines whether individual interventions are reproducibly separable, and benchmark-level ranking resolution determines whether the perturbation set as a whole can recover differences among predictors; the two can diverge, because a large screen can pool many individually unresolved perturbations into a benchmark that still orders graded predictors correctly. This applies beyond coding variants to related drugs, doses, combinations, weak genetic perturbations and subtle cell-state transitions.

Experimental design and computational evaluation are therefore coupled rather than sequential concerns. Whether sibling alleles are resolvable is set jointly by effect magnitude, inter-allele separation, biological heterogeneity, technical noise, variant assignment quality, cellular context and stimulation, sampling time point and readout modality, not by nominal cell depth alone. Deeper profiling is consequently not always sufficient. A small pilot can separate perturbations likely to gain from additional cells from those for which depth alone may not deliver resolution, but that triage is a detection-level judgement about a perturbation against its reference, not a guarantee that sibling alleles will become

identifiable. Where depth is not the binding constraint, allele-resolved studies may need more informative designs: biological replication, improved variant assignment, matched stimulation, responsive cell states, time-resolved sampling, or orthogonal readouts of chromatin state, protein abundance, signaling activity and phenotype. Model design faces the mirror-image constraint. More expressive protein representations cannot compensate for an unresolved transcriptional target, and conversely a resolvable target does not guarantee that current sequence- or feature-based models capture the context-dependent mapping from molecular variation to cellular response. Allele-conditioned models will need to connect protein structure and biochemical mechanism with pathway state and cellular context, and they will need to be evaluated in datasets whose allele-specific responses are independently reproducible.

Several limitations bound the scope of these conclusions. The publicly available allele-resolved datasets cover only four genes, spanning three technologies and three cell contexts, so gene identity is partly confounded with measurement regime and cellular context. The observed differences must therefore be read as dataset-specific measurement regimes, not as intrinsic rankings of allele biology. The model-limited verdict also rests on a comparatively small JAK1 panel: 26 variants in total, 20 entering the native rankability analysis, 14 at the matched 50-cell depth, and 9, 6 and one variant at the successively deeper titration levels. The prospective pilot analysis predicts detection-level rankability, that is, resolution of a perturbation against its reference, which is necessary but not sufficient for a power calculation on within-gene allele identification. The replicate-based ceiling is an empirical reproducibility reference for a specified depth and scoring protocol, not a mathematical upper bound on all denoising or hierarchical models. Pseudobulk profiles are the prediction target throughout, and although cell-level classifier controls show that the identification floor is not an artifact of averaging, pseudobulk evaluation may still underrepresent effects confined to rare subpopulations or shifts in cell-state composition. Finally, transcriptomic response is not the whole phenotype: alleles that are indistinguishable in expression space may still differ in protein state, signaling activity or organismal consequence. Controlled studies profiling multiple genes within shared experimental contexts, with biological replicates, depth titrations and multimodal readouts, will be needed to establish identification-aware power criteria.

Read together, these results support a measurement-aware principle for predictive cellular modeling: model performance is interpretable only at biological resolutions that the experiment itself can reproduce. What allele-resolved modeling is asked to do should change accordingly. Predictors should be judged on whether they recover allele-specific residual structure, not the gene-shared response that dominates direction-based scores, and allele-resolved experiments should establish identification before prediction is scored at all. The same requirement extends beyond coding variants to drug analogues, dose series, combinatorial perturbations and multimodal phenotypes, wherever closely related interventions must be told apart rather than merely detected. Progress in variant-aware virtual-cell modeling will therefore not depend on more expressive architectures alone; it will depend equally on measurements and benchmarks that preserve the distinctions those architectures are asked to predict. By separating what is shared across variants from what identifies an allele, and what is unmeasured from what remains unmodeled, this study provides a foundation for deciding when allele-specific cellular prediction is scientifically testable, and for building allele-resolved virtual-cell models on experimentally resolvable ground truth.

Methods

Study design

This study is a retrospective evaluation of held-out protein-coding variant prediction from single-cell perturbation data, with AllelePerturb as its task formalization and evaluation protocol. All data are publicly available single-cell perturbation datasets, and no new human, animal or biological samples were generated. The primary unit of prediction and of statistical resampling is the protein-coding variant condition. Throughout, *allele* denotes this experimentally assigned protein-coding variant condition rather than a phased genomic haplotype.

Four genes (TP53, KRAS, GATA1 and JAK1) assayed by three technologies (Perturb-seq, base editing and scSNV-seq) were included. To our knowledge, these are the publicly available single-cell perturbation datasets that resolve individual protein-coding variants with reusable per-variant labels. All eligible

variants and cells were used, so sample size was not predetermined.

For each generalization split, models were trained on the training variants of a gene and evaluated on its held-out variants. Every model was scored through a single evaluation harness. The primary analyses quantify transcriptional direction recovery and within-gene allele discrimination. Separate replicate-based analyses assess perturbation-versus-wild-type detection, within-gene allele identification and benchmark-level model-ranking resolution. Sensitivity analyses varied the expression representation, distance metric, sampling depth and preprocessing protocol.

Benchmark construction

Data sources. AllelePerturb-Bench integrates single-cell perturbation data for four protein-coding genes assayed across three experimental technologies. TP53 (98 variants) and KRAS (92 variants) were obtained from a Perturb-seq screen of coding variants in A549 lung adenocarcinoma cells (Gene Expression Omnibus GSE161824)²³. GATA1 (254 variants) was obtained from a base-editing screen in human hematopoietic stem and progenitor cells. For that gene we used the processed expression matrix and predefined holdout splits distributed with PerturbNet²⁴ (Hugging Face repository cyclopet/perturbnet_reproduce). JAK1 (26 variants) was obtained from a single-cell single-nucleotide-variant sequencing (scSNV-seq) experiment in HT-29 colorectal cancer cells stimulated with interferon-gamma (European Nucleotide Archive PRJEB48915; processed objects from Zenodo record 10418435)²⁵. Cell counts include both variant-assigned cells and the dataset-specific wild-type or control cells retained after preprocessing. The four datasets together comprise 470 protein-coding variants and 321,043 cells.

Preprocessing. For TP53 and KRAS, processed count matrices, cell barcodes and variant-to-cell assignment tables were loaded from GEO. Cells were assigned to variants using the published assignment files, giving a shared 939-gene expression space for the two A549 datasets. For GATA1, we used the PerturbNet-distributed AnnData object. It provides log-normalized expression over 2,477 highly variable genes together with five predefined holdout partitions. For JAK1, the SingleCellExperiment object was loaded in R (Bioconductor). Cells were filtered to homozygous genotyped calls with unambiguous coding consequences (missense, splice-site, stop-gained or synonymous). The 2,000 most highly variable genes were selected with `scrn::modelGeneVar` followed by `getTopHVGs`, and log-normalized counts were exported.

Each dataset was processed in its own gene space, because the three cell types do not share a common highly-variable-gene set. The object shared across datasets is the variant feature representation, not the gene space. Within each dataset, every gene was standardized to zero mean and unit variance across all cells, and wild-type cells served as the common reference.

This standardization is transductive. Refitting it on training variants and wild-type cells only, excluding held-out variants, left every predictor at or below chance (PDS-cosine ≤ 0.48 , no predictor above 0.50). Direction recovery was essentially unchanged, and model scores were if anything lowered (mean $\Delta\text{PDS} = -0.01$). The reported all-cell standardization is therefore conservative with respect to the null (Supplementary Note).

Variant feature vector. Each variant v was represented by protein-level features rather than by a variant identifier, so that held-out variants unseen in training can still be predicted. The primary representation is a six-dimensional biophysical vector

$$\theta_v = (\Delta H_v, \Delta V_v, \Delta Q_v, f_v, s_v, h_v),$$

where the first three components are the mutant-minus-wild-type differences in Kyte-Doolittle hydrophobicity, residue molecular weight and formal charge at pH 7.0. Write $p(\cdot)$ for a residue property, and $a_v^{\text{mut}}, a_v^{\text{wt}}$ for the mutant and wild-type amino acids of variant v . Each biophysical difference is then $\Delta p_v = p(a_v^{\text{mut}}) - p(a_v^{\text{wt}})$, standardized across all 470 variants by $z(\Delta p_v) = (\Delta p_v - \mu_{\Delta p}) / \sigma_{\Delta p}$.

The remaining three components are structural indicators. The fold-core term f_v was set to 1.0 for residues within an annotated structural core, to 0.5 for annotated interaction regions, and to 0.0 elsewhere. The annotated cores are the DNA-binding, G-, zinc-finger and kinase domains of TP53, KRAS, GATA1 and JAK1, respectively. The functional-switch term s_v was set to 1.0 for catalytic or metal-

coordinating residues. The hotspot or pathogenic marker h_v was defined exclusively from external prior knowledge, with no reference to the single-cell outcome. Its sources were canonical COSMIC and IARC missense hotspot codons, ClinVar pathogenic annotations and gene-specific functional-module membership. The full per-gene residue and codon assignments are listed in Supplementary Table 1.

An earlier version of h_v used the top quintile of measured effect size, which leaks the evaluation outcome into the feature. We removed it and re-ran the entire evaluation grid. The ranking result was unchanged after this correction: θ -only PDS shifted by at most 0.02, and every method remained at chance (Supplementary Information).

Protein-language-model embeddings. As an alternative representation, we extracted per-variant embeddings from ESM-1v (esm1v_t33_650M_UR90S_1)³³. Write $\mathbf{H}_v \in \mathbb{R}^{L_v \times 1280}$ for the final-layer hidden states of the full-length mutant sequence of length L_v . The embedding is the mean over residue positions,

$$e_v = \frac{1}{L_v} \sum_{\ell=1}^{L_v} \mathbf{H}_v[\ell, :] \in \mathbb{R}^{1280},$$

used directly, without dimensionality reduction, as the ESM feature space.

Evaluation protocol (AllelePerturb-Eval)

Pseudobulk perturbation profiles. Let C_v and C_{WT} denote the cells of variant v and of wild type. Let $x_i \in \mathbb{R}^G$ denote the standardized expression of cell i over G genes. The perturbation profile of variant v is the pseudobulk difference

$$\delta_v = \frac{1}{|C_v|} \sum_{i \in C_v} x_i - \frac{1}{|C_{WT}|} \sum_{j \in C_{WT}} x_j.$$

To bound the influence of the most deeply sampled variants, up to 300 cells were sampled without replacement per variant and per wild-type reference. Variants with fewer than five cells were excluded. Every predictor maps a variant feature vector to an estimate $\hat{\delta}_v$ of this profile, and all metrics compare $\hat{\delta}_v$ to the measured δ_v . The prediction target throughout is this population-average profile. The split-half distributional analyses introduced below are diagnostics of measurement resolution on the single-cell distributions, not alternative prediction targets.

AllelePerturb-Eval separates prediction quality into three families of metrics. **Direction recovery** asks whether a predicted profile points the right way in expression space. **Allele discrimination** asks whether a predicted profile is specific enough to identify its own variant among the other held-out variants. **Differential-expression fidelity** asks whether the predicted and measured profiles agree on which genes change and by how much. The ten metrics are defined below. Unless noted, each is computed per held-out variant and then averaged over the held-out set T of a gene, over genes and over splits, where $n_{\text{test}} = |T|$.

Perturbation discrimination score (PDS). PDS is the primary metric and directly tests allele discrimination. It asks whether a predicted profile is closer to its own measured profile than to those of the other held-out variants. This nearest-match discrimination logic follows the perturbation-discrimination principle used in recent virtual-cell evaluation efforts^{3,27}. We adapt that principle to protein-coding variants and compute it with tie-aware ranking.

For a held-out variant v of gene g , the predicted profile $\hat{\delta}_v$ is ranked by distance $d(\cdot, \cdot)$ against the measured profiles of all candidate variants of the same gene. The candidate set is $C_g = \text{train}(g) \cup \text{test}(g)$, of size $n = |C_g|$, and it holds both training and held-out variants passing the five-cell filter. Let $\ell_v = |\{w \in C_g : d(\hat{\delta}_v, \delta_w) < d(\hat{\delta}_v, \delta_v)\}|$ be the number of candidates strictly closer than the true match. Let $e_v = |\{w \in C_g : d(\hat{\delta}_v, \delta_w) = d(\hat{\delta}_v, \delta_v)\}| \geq 1$ be the number tied with it, including v itself. The tie-aware score is

$$\text{PDS}(v) = 1 - \frac{\ell_v + \frac{1}{2}(e_v - 1)}{n - 1},$$

so the nearest match scores 1 and random matching 0.5 in expectation, with ties assigned the average (mid) rank. Only held-out variants are scored, but each competes against the full same-gene candidate set. A correct prediction must therefore be closer to its own variant than to every other measured variant of the gene, held out or training. A prediction with $\|\hat{\delta}_v\| = 0$, or a gene with a single candidate ($n \leq 1$), is assigned 0.5 by definition.

PDS was computed under three distances,

$$d_{\cos}(a, b) = 1 - \frac{a^\top b}{\|a\| \|b\|}, \quad d_1(a, b) = \|a - b\|_1, \quad d_2(a, b) = \|a - b\|_2,$$

giving PDS-cosine (primary), PDS-L1 and PDS-L2. The WT-null reference predicts $\hat{\delta}_v = 0$ for every variant, and therefore scores exactly 0.5 by the zero-prediction rule above. Its cosine distance to a zero vector is undefined and is not evaluated. The empirical chance level under small, gene-stratified candidate sets with ties can depart slightly from the analytic 0.5, so we additionally estimated it by label permutation (see Statistical analysis).

Direction-recovery metrics. Averaging the Pearson correlation over held-out variants gives

$$\text{Pearson-}\delta = \frac{1}{n_{\text{test}}} \sum_{v \in T} \text{corr}(\hat{\delta}_v, \delta_v),$$

and δ -cosine is the analogous average of $\hat{\delta}_v^\top \delta_v / (\|\hat{\delta}_v\| \|\delta_v\|)$. Pearson- δ (top-20) restricts the correlation to the 20 genes with the highest variance across the training-set profiles. This restricted version tests whether agreement is concentrated in a small number of high-variance genes.

Differential-expression metrics. For each variant, differentially expressed genes were defined by a per-gene Welch t -test of variant against wild-type cells (up to 300 cells per group). Let $\mathcal{D}_v^{\text{true}}$ be the $k = 50$ genes with the largest absolute t -statistic, and $\mathcal{D}_v^{\text{pred}}$ the k genes with the largest absolute predicted change. DE overlap is

$$\text{DE overlap}(v) = \frac{|\mathcal{D}_v^{\text{pred}} \cap \mathcal{D}_v^{\text{true}}|}{k}.$$

DE-LFC Spearman is the Spearman correlation between the measured log-fold-changes $\ell_{v,g}$ and the predicted changes $\hat{\delta}_{v,g}$ over $\mathcal{D}_v^{\text{true}}$. Direction agreement is the fraction of genes in $\mathcal{D}_v^{\text{true}}$ for which $\text{sign}(\hat{\delta}_{v,g}) = \text{sign}(\ell_{v,g})$. Both quantities use the Welch-test log-fold-change $\ell_{v,g}$ from the differential-expression subsample as the measured reference, computed independently of the pseudobulk profile δ_v .

Reconstruction metric. Mean absolute error is $\text{MAE}(v) = G^{-1} \sum_g |\hat{\delta}_{v,g} - \delta_{v,g}|$, averaged over held-out variants.

Generalization splits

Six split strategies probe distinct axes of generalization. In each, models are trained on the training variants of a gene and evaluated on its held-out variants.

1. Random: the predefined per-dataset holdout partition shipped with AllelePerturb-Bench, with GATA1 using the PerturbNet partition. The realized held-out sets are 16/98 (TP53), 8/92 (KRAS), 52/254 (GATA1) and 9/26 (JAK1) variants. This approximates a random holdout, but the per-gene fraction varies with the source partition rather than being fixed at a single value.
2. Positional (out-of-distribution position): variants in the C-terminal half of each protein held out.
3. Mechanistic (out-of-distribution mechanism): variants at hotspot or functional-switch residues held out.
4. Cross-gene: all GATA1 and JAK1 variants held out, with training on the TP53 and KRAS variants. This split is defined for completeness but excluded from the main grid, because the datasets do not share a gene space. Cross-gene transfer is therefore not defined for a shared decoder (see Discussion).

5. Low-depth: variants with fewer than 200 cells held out, isolating the shallow-sampling regime. Only GATA1 (45 variants) and JAK1 (17) contain such variants, because all TP53 and KRAS variants exceed this depth.
6. Compatibility: the GATA1 held-out set matched to the PerturbNet evaluation protocol, that is, the same 52 variants as the random split. It is combined with the predefined TP53 (10) and KRAS (8) holdouts, enabling comparison with prior work.

Prediction methods

We evaluated 20 predictors: 18 feature-model combinations formed by crossing six regression heads with three feature spaces, plus two references. Each predictor learns a map $g_\phi : \mathbb{R}^p \rightarrow \mathbb{R}^G$ from a p -dimensional variant feature vector to a G -dimensional pseudobulk profile. Fitting minimizes squared error on the training variants, $\min_\phi \sum_{v \in \text{train}} \|g_\phi(\text{feat}_v) - \delta_v\|_2^2$, with the head-specific regularizer below. Every predictor was scored by the identical protocol above.

Regression heads. We used six standard regression heads spanning linear, tree-ensemble, neighbor and neural-network families. These are ridge, lasso, random forest, gradient-boosted trees, k -nearest-neighbors regression and a multilayer perceptron (hyperparameters in Supplementary Table 2). Linear and neighbor heads regress directly onto the full expression profile. Tree ensembles scale poorly to thousands of output genes. For output spaces larger than 200 genes, the random forest and gradient-boosted heads were therefore fit on a q -component PCA of the training profiles. Predictions were mapped back to gene space by the inverse transform. All heads used a fixed random seed.

Feature spaces. The feature spaces are θ (six-dimensional biophysical vector), ESM (1,280-dimensional ESM-1v embedding) and ESM+ θ (their 1,286-dimensional concatenation). As a representation robustness control (Supplementary Note) we additionally extracted ESM2-650M²⁸ final-layer representations of the mutant sequence, three mutation-aware constructions of the mutant-minus-wild-type difference, and AlphaMissense²⁹ pathogenicity scores. The three constructions are the global mean difference, the mutated-residue hidden-state difference $h_p^{\text{mut}} - h_p^{\text{wt}}$, and a ± 16 -residue local-window difference. Proteins longer than 1,022 residues were cropped to a window centered on the mutation, and multi-substitution variants averaged their site differences. All were scored through the identical splits, heads and metrics.

References. The WT-null reference predicts $\hat{\delta}_v = 0$ for every variant, quantifying the score attainable with no perturbation signal. The Gene-mean reference predicts the mean training-variant profile of the held-out variant’s own gene, $\hat{\delta}_v = |\text{train}(g_v)|^{-1} \sum_{w \in \text{train}(g_v)} \delta_w$. Because it uses the gene label g_v , it has access to information a variant-blind predictor does not. We therefore report it as a gene-identity reference on how much of the response is explained by gene identity alone, rather than as a competing method. Gene-mean is defined only for genes with at least one training variant in a split, and is therefore omitted under the cross-gene split.

Published models under a unified evaluation. To compare against published perturbation-prediction methods, every model was scored through a single evaluation harness. That harness uses the same expression arrays, deterministic pseudobulk profiles, tie-aware PDS and bootstrap and permutation procedures as the in-house heads, across the five main splits (Supplementary Table 4). Each model was trained per gene on the training variants and their wild-type control, and predicted each held-out variant from its feature vector. Held-out variant response cells were never used in training.

The published scGen and scVIDR packages condition on a categorical perturbation label rather than a continuous per-variant feature. We therefore used in-house re-implementations of their latent vector arithmetic, following refs^{6,34}. Each re-implementation is a per-(gene, split) variational autoencoder with θ as the perturbation attribute, and the predicted latent shift is decoded onto wild-type cells. scVIDR additionally scales that shift by a per-variant severity term. Biolord³⁵ (official biolord 0.0.3, ordered-attribute disentanglement) and CellFlow³⁶ (official cellflow-tools, conditional optimal-transport flow) were run from their released packages. Each variant’s ESM-1v embedding served as the CellFlow condition, including for held-out variants, so unseen alleles resolve by feature.

We additionally adapted CPA⁷ (official cpa-tools 0.8.8) in a chemCPA style, replacing its categorical perturbation dictionary with a frozen encoding of the variant features. This model produced non-finite

training losses on the continuous inputs and did not converge, even with gradient clipping and reduced learning rates. It is reported for interface completeness rather than scored. Three further models (scGPT¹⁰, GEARS⁹ and STATE³⁷) condition on perturbed-gene identity and assign an identical prediction to every variant of a gene. For these we report allele coverage (4 of 470 variants) rather than a discrimination score, which is undefined for a gene-invariant predictor.

PerturbNet²⁴ (conditioned on ESM-1v) generates within a 50-dimensional principal-component subspace fit on wild-type cells. Random predictions confined to this subspace already score above the analytic chance level. We therefore evaluated PerturbNet against a permutation null estimated in the same subspace, under which its exceedance was not significant. Subspace-inflation controls and values are given in Supplementary Table 4 and results/canonical/perturbnet_subspace_test.json. Confidence intervals are 2,000-sample bootstraps over held-out variants, and permutation nulls use 1,000 prediction-to-target shuffles, as for the in-house grid.

Split-half detection analysis

To separate model limitation from measurement limitation, we asked whether variant responses are stably measurable at the available sampling depth. The energy distance between two cell sets $X = \{x_i\}_{i=1}^n$ and $Y = \{y_j\}_{j=1}^m$ is

$$E(X, Y) = \frac{2}{nm} \sum_{i=1}^n \sum_{j=1}^m \|x_i - y_j\| - \frac{1}{n^2} \sum_{i,i'} \|x_i - x_{i'}\| - \frac{1}{m^2} \sum_{j,j'} \|y_j - y_{j'}\|,$$

with Euclidean norms. For each variant with at least 50 cells, cells were partitioned into two random halves A and B , and a size-matched wild-type control sample C was drawn. We computed the within-variant replicate distance $D_{\text{self}} = E(A, B)$ and the variant-to-wild-type distance $D_{\text{null}} = E(A, C)$.

As a control on the reference, we additionally computed the wild-type-to-wild-type distance D_{wtwt} from an independent split of the wild-type cells. For TP53, KRAS and GATA1 this equaled D_{null} ($D_{\text{wtwt}}/D_{\text{null}} = 0.91\text{--}1.09$), so the detection floor reflects the absence of variant signal rather than control heterogeneity. For JAK1, by contrast, D_{null} exceeded D_{wtwt} eight- to thirty-fold (Supplementary Note).

Distances were evaluated over a range of subsample depths per half and in three representation spaces, with 50 random seeds per configuration. The canonical values use a 50-component PCA at native maximum depth (Supplementary Table 2). Each seed distribution was summarized by its median and its 2.5 to 97.5 percentile interval.

A variant was classified as detection-eligible (labeled *rankable* in the figures) when its variant-to-wild-type signal exceeded the sampling width of its replicate distance,

$$S = \text{med}(D_{\text{null}}) - \text{med}(D_{\text{self}}) > W = Q_{97.5}(D_{\text{self}}) - Q_{2.5}(D_{\text{self}}),$$

and un-rankable otherwise. This signal-versus-noise criterion is the primary definition. We confirmed that the conclusions are qualitatively unchanged under four alternatives: a lower confidence bound of S above zero, $S > 1.5W$, $S > 2W$, and a ratio-based threshold. These alternatives shift absolute fractions but preserve the ordering across datasets (Supplementary Fig. 1).

The same analysis was applied to the four AllelePerturb genes and to three widely used gene-level Perturb-seq atlases. Those atlases are Replogle 2022⁴, Norman 2019³² and Adamson 2016³¹ (K562 cells), obtained from the scPerturb data portal¹⁸. An identical native-depth, all-perturbation, direct split-half definition was used throughout. Here native depth denotes each perturbation evaluated at the deepest split-half subsample rung it supports, from 25 to 800 cells per half, in the 50-component PCA space. Shallower datasets are therefore scored at correspondingly shallower depth. The resulting per-dataset rankability results are those reported in Fig. 4h (rankable fraction) and Fig. 6f (un-rankable fraction).

Rankability predictor

To test whether detection-eligibility can be anticipated before a full model comparison, we trained a logistic-regression classifier to predict per-perturbation rankability under leave-one-dataset-out cross-validation. The target is the detection-level split-half label defined above. Each perturbation contributed

a single example: its canonical native-depth rankability label (deepest available split-half bin, energy distance, PCA space, as above) and its pilot-estimable pseudobulk effect size $\|\delta_v\|_2$.

We excluded the split-half quantities that define the label (S , W , their ratio, D_{self} , D_{null}) and any evaluation-configuration identity (distance metric, representation space, subsample depth). These quantities are not properties of the perturbation and would leak the label. Discrimination was measured by the area under the receiver-operating-characteristic curve, with 2,000-sample bootstrap intervals over held-out perturbations. Held-out datasets whose label was single-class, namely TP53 and KRAS at native depth (all un-rankable), are reported as not evaluable rather than assigned an undefined AUROC. The predictor is implemented in `scripts/figures/rankability_predictor.py`.

Prospective pilot-to-full validation. To test that the rankability predictor is prospective rather than a retrospective consequence of same-source data, we re-evaluated it on cells disjoint from those defining the label. For each perturbation, a single random permutation assigned the first 50 cells to a pilot set and the next $2T$ cells to an evaluation set. The pilot effect size $\|\hat{\delta}\|_2$, and a pilot signal-to-noise ratio from a pilot split-half, were computed from 25 or 50 pilot cells. The rankability label at target depth $T \in \{100, 200\}$ was computed by the split-half criterion on the disjoint evaluation cells.

A leave-one-dataset-out logistic predictor and a training-free pilot signal-to-noise threshold were evaluated across the seven datasets. Because the rankability label uses only evaluation cells disjoint from the cells that form the pilot features, this is a prospective test. We report per-dataset AUROC over the datasets with balanced labels rather than a pooled value, because pooling mixes differing per-dataset base rates. A within-dataset label-permutation control returned a null AUROC of 0.49. Scripts are provided under `results/pilot_validation/`.

Measurement ceiling, pairwise identification and benchmark resolution

Three analyses separate measurement limits from model limits. All reuse the identical deterministic pseudobulk and tie-aware PDS of the evaluation protocol, average over 15 random cell subsamples, and require no additional data.

Oracle ceiling. For each variant with at least five cells, cells were partitioned into two disjoint halves, and a size-matched wild-type control was split likewise, giving two independent pseudobulk deltas. Using one half as the prediction and the other as the evaluation truth, we scored PDS through the identical harness. That harness used the same candidate sets, tie-aware ranking and split structure as the model comparison. Because the prediction is an independent second measurement of the same variant, its PDS defines an empirical replicate-based discrimination ceiling at that depth and protocol. We report it as an empirical, depth- and protocol-specific quantity, rather than a mathematical upper bound on models that borrow information across variants. PerturbNet is excluded from this per-gene comparison, because it is scored in a reduced principal-component subspace that is not comparable to the full-gene-space oracle.

Ceiling sensitivity and gene-shared decomposition. We assessed the sensitivity of the ceiling to the split-half protocol. Recomputing the oracle using all available cells per half rather than the 300-cell cap left it unchanged. We also substituted a near-true prediction: the deep mean of a variant, taken from cells disjoint from the evaluation truth. Scored against the benchmark’s noisy pseudobulk truth, this near-true prediction remained at chance for the low-effect genes. Shrinking predictions toward the gene mean did not raise discrimination. These sensitivity outcomes are reported with the corresponding results (Supplementary Note).

Separately, we decomposed each effect into a gene-shared mean and an allele-specific residual, $\delta_v = \bar{\delta}_g + r_v$. We then recomputed the oracle and the per-predictor Pearson- δ on the residual profiles, to quantify how much of direction recovery is gene-shared.

Pairwise identification. In the same cosine-delta space, we computed the split-half replicate distance $D_{\text{self}}(v)$ and the pairwise distance $D(v_i, v_j)$ between variant mean deltas at a matched half-depth. A pair was called resolvable when $D(v_i, v_j)$ exceeded the mean of the two variants’ replicate distances. A variant was called identifiable when its nearest sibling lay beyond its own replicate noise.

Metric-family and classifier controls. To test whether the result depends on the energy-distance metric, we recomputed the split-half window ratio $D_{\text{self}}/D_{\text{null}}$ in the 50-component PCA space under five distances. These were energy, maximum-mean-discrepancy (RBF kernel, median-heuristic bandwidth), sliced-Wasserstein, cosine and Euclidean distances (Supplementary Note). As a distribution-free

identification test, we trained an ℓ_2 -regularized logistic-regression classifier to separate the single cells of two conditions under five-fold cross-validation in the same space. The principal components were fit label-blind, and classes were balanced by subsampling. We report the area under the ROC curve for variant-versus-variant (identification) and variant-versus-wild-type (detection) comparisons.

Because all variants of a gene derive from one pooled experiment, there is no per-variant batch structure. Label-permutation and wild-type-versus-wild-type controls both returned a chance AUROC of 0.5. Where the classifier exceeded chance, its accuracy survived regressing out per-cell depth.

Benchmark resolution. To test whether the oracle ceiling governs benchmark validity, we built synthetic predictors of known, graded quality, $\hat{\delta}(v, \alpha) = (1 - \alpha) \bar{\delta}_g + \alpha \delta_v^{\text{build}}$. Here δ_v^{build} is a pseudobulk delta from a build subsample disjoint from the evaluation truth, and $\bar{\delta}_g$ is the gene mean. True quality increases monotonically with α , and at $\alpha = 1$ the attainable score equals the oracle ceiling. Scoring the family through the harness and bootstrapping the held-out variants, we recorded the probability of recovering the full α -ordering and of selecting the top predictor. Both probabilities were recorded as a function of the oracle window and of the number of held-out variants.

External benchmark calibration. We applied the identical controlled-predictor resolution analysis to five public perturbation datasets: Norman, Replogle, Adamson, the Virtual Cell Challenge training set and sci-Plex³⁰. For each we loaded the ground-truth single-cell profiles and computed the resolution, that is, the probability of recovering the true predictor ordering under bootstrap over perturbations. This was done at 50 cells per half, over perturbations with at least 100 cells (Fig. 5f). Because this inclusion threshold and depth differ from the native-depth main analysis, the allele perturbation counts and rankable fractions there differ from Supplementary Table 5. Analysis scripts are provided under `results/benchmark_resolution/`.

Analytical scaling. Under a finite-sample pseudobulk model, two half-profiles of a variant satisfy $\mathbb{E}\|d_A - d_B\|^2 = 2\eta^2$, with per-profile noise $\eta^2 = \text{tr}(\Sigma)/m$. Two variants satisfy $\mathbb{E}\|d_i - d_j\|^2 = \|\mu_i - \mu_j\|^2 + 2\eta^2$. Together these give $(D_{\text{self}}/D_{\text{null}})^2 = 1/(1 + S)$ with $S = \|\delta\|^2/(2\eta^2)$, and a dimensionless discrimination signal-to-noise $\rho = \bar{\Delta}\sqrt{m}/(\sqrt{2}\eta)$. Discrimination depends on the data only through ρ and the number of candidate variants. We confirmed the derived depth scaling empirically ($\eta^2 \propto 1/m$ across all genes). The noise-debiased between-variant signal $\bar{\Delta}^2$ lay at or below the sampling-noise floor for TP53, KRAS and GATA1 at every tested depth, so only JAK1 traced the increasing branch of ρ . The scaling is derived; the functional form is calibrated from the synthetic sweep rather than assumed for the tie-aware cosine score.

Power analysis

To relate detection to sampling depth, cells were subsampled at 50, 100, 150 and 300 per variant, and the split-half rankability test was recomputed at each depth. The fraction of variants classified as detectable was plotted against subsample size for each gene. A detection threshold, the depth at 50% detection, was estimated by sigmoid fitting where the data supported it.

Statistical analysis

Confidence intervals for PDS and Pearson- δ were obtained by nonparametric bootstrap over held-out variants (2,000 resamples, percentile method, 95% level). A method's PDS was called to differ from chance when its 95% confidence interval excluded 0.50. Its direction was called significantly positive when the lower bound of its Pearson- δ interval exceeded 0. The per-gene $D_{\text{self}}/D_{\text{null}}$ ratio (Fig. 4c) was bootstrapped over variants. Rankability fractions are binomial proportions over perturbations and carry Wilson 95% score intervals (Fig. 6f); the two depth arms of Fig. 6f are computed over the perturbations measured at both depths, so the comparison is paired. Shaded bands in Fig. 5f,g and Fig. 4c are reading aids with the numeric edges given in those captions, and are not used as thresholds in any analysis.

To confirm that the analytic PDS chance level of 0.50 holds under the actual tie structure and small test sets, we ran a within-split label-permutation control. For each method-split-gene combination, the full prediction-to-truth distance matrix was precomputed. The prediction-to-target correspondence was then permuted 1,000 times, recomputing PDS with the same tie-aware ranking each time. The empirical null mean was 0.500, with a range of 0.495 to 0.504 across 340 combinations. Only 3.2% of combinations exceeded the permutation 95th percentile, consistent with chance under the empirical permutation dis-

tribution. Because these method-split-gene combinations are correlated, this is a descriptive check rather than a formal false-positive calibration. All bootstraps and permutations used fixed random seeds.

Because some residues carry more than one substitution, we also resampled residue-level clusters rather than individual variants. This left the random-split intervals essentially unchanged: 23 of 24 predictors still overlapped chance, with PerturbNet as the sole exception; its apparent exceedance was not significant against the permutation null estimated in its 50-dimensional output subspace. Training-to-held-out residue overlap ranged from 0% (positional split) to 54% (compatibility split) without altering the chance-level result (Supplementary Note).

We report intervals to characterize uncertainty rather than as per-method hypothesis tests. The conclusions rest on the direction and consistency of the effect across all 19 non-null predictors, which move together. They therefore do not depend on any single comparison or on a multiplicity correction. These intervals quantify within-benchmark uncertainty over held-out variants, not generalization across the four genes, assays or technologies, which the four-dataset design cannot estimate.

To ensure that discrimination estimates did not depend on a single stochastic pseudobulk draw, each variant's PDS in the method comparison was averaged over 15 independent random subsamples of cells. The reported point estimate is then the mean over (split, gene) cells of the within-cell mean of these seed-averaged per-variant scores. The 95% confidence interval is a 2,000-sample nonparametric bootstrap that resamples held-out variants within each (split, gene) cell. No statistical method was used to predetermine sample size, and all eligible public variants and cells were included. Measurement power was instead assessed retrospectively and prospectively through the split-half and pilot analyses described above.

Software and reproducibility

The evaluation grid was run in Python 3.11.14 with NumPy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2 and SciPy 1.14.1. Single-cell loading and preprocessing used scanpy 1.10.3 (Python) and scan with Single-CellExperiment (R, Bioconductor). ESM-1v embeddings were extracted with fair-esm 2.0.0 and PyTorch 2.10.0 (CUDA 12.6). Figures were rendered with matplotlib 3.10.8. Published-model adapters were run in their own package environments. Biolord 0.0.3 and the scGen/scVIDR re-implementations ran under scvi-tools 1.4.3 with PyTorch 2.12. CellFlow (cellflow-tools 0.1.1) ran under JAX, and CPA (cpa-tools 0.8.8) ran under scvi-tools 0.20.3 with PyTorch 2.0.

Data availability

All single-cell datasets analyzed in this study are publicly available. The TP53 and KRAS Perturb-seq data are available from the Gene Expression Omnibus under accession GSE161824²³. The GATA1 base-editing data were originally generated by Perturb(BE)-seq in human hematopoietic stem and progenitor cells, and deposited at the Gene Expression Omnibus under accession GSE215253. We used the processed expression matrix and predefined holdout splits redistributed with PerturbNet (Hugging Face repository `cyclopet/PerturbNet_reproduce`)²⁴. The JAK1 scSNV-seq data are available from the European Nucleotide Archive under accession PRJEB48915, with processed objects at Zenodo (<https://doi.org/10.5281/zenodo.10418435>)²⁵. The gene-level Perturb-seq atlases used for the rankability analysis (Replogle 2022, Norman 2019 and Adamson 2016) were obtained from the scPerturb data portal¹⁸. The derived AllelePerturb-Bench tables that support the main figures and the Supplementary Information are available at <https://github.com/Boom5426/AllelePerturb>. These tables comprise the per-variant feature vectors, cell counts and generalization-split assignments, together with the per-variant and per-method evaluation grids. The per-gene expression arrays and the pseudobulk profiles derived from them are not redistributed; they can be reconstructed from the public accessions above using the preprocessing described in Benchmark construction.

Code availability

Custom code and the committed result tables are available at <https://github.com/Boom5426/AllelePerturb>. Each canonical table is annotated with the panels and Supplementary values it supports. Analyses that read only committed tables, including the rankability predictor, the pilot validation and the residue-independence control, can be run from a fresh checkout. Other analyses

require processed per-gene expression arrays, protein embeddings, external single-cell atlases or a shared evaluation harness from the original compute environment. These data dependencies are not redistributed where restricted by size or source availability, and the shared harness and the adapters used to run the published models are not included in the current repository. The relevant scripts accept the compute-workspace location as a parameter, report the full expected path of missing inputs and document the required directory layout in the repository. Random seeds are fixed throughout (Statistical analysis).

Reporting summary

References

- [1] Evan H. Baugh, Hua Ke, Arnold J. Levine, Richard A. Bonneau, and Chris Sander Chan. Why are there hotspot mutations in the TP53 gene in human cancers? *Cell Death & Differentiation*, 25:154–160, 2018. doi: 10.1038/cdd.2017.180.
- [2] John C. Hunter, Arianna Manandhar, Mario A. Carrasco, et al. Biochemical and structural analysis of common cancer-associated KRAS mutations. *Molecular Cancer Research*, 13:1325–1335, 2015. doi: 10.1158/1541-7786.MCR-15-0203.
- [3] Charlotte Bunne, Yusuf Roohani, Yanay Rosen, et al. How to build the virtual cell with artificial intelligence: Priorities and opportunities. *Cell*, 187:7045–7063, 2024. doi: 10.1016/j.cell.2024.11.015.
- [4] Joseph M. Replogle, Reuben A. Saunders, Angela N. Pogson, et al. Mapping information-rich genotype-phenotype landscapes with genome-scale perturb-seq. *Cell*, 185:2559–2575.e28, 2022. doi: 10.1016/j.cell.2022.05.013.
- [5] Jesse Zhang, Airol A Ubas, Richard de Borja, et al. Tahoe-100m: A giga-scale single-cell perturbation atlas for context-dependent gene function and cellular modeling. *bioRxiv*, 2025. doi: 10.1101/2025.02.20.639398. Preprint.
- [6] Mohammad Lotfollahi, F. Alexander Wolf, and Fabian J. Theis. scGen predicts single-cell perturbation responses. *Nature Methods*, 16:715–721, 2019. doi: 10.1038/s41592-019-0494-8.
- [7] Mohammad Lotfollahi, Anna Klimovskaia Susmelj, Carlo De Donno, et al. Predicting cellular responses to complex perturbations in high-throughput screens. *Molecular Systems Biology*, 19:e11517, 2023. doi: 10.15252/msb.202211517.
- [8] Charlotte Bunne, Stefan G. Stark, Gabriele Gut, et al. Learning single-cell perturbation responses using neural optimal transport. *Nature Methods*, 20:1759–1768, 2023. doi: 10.1038/s41592-023-01969-x.
- [9] Yusuf Roohani, Kexin Huang, and Jure Leskovec. Predicting transcriptional outcomes of novel multigene perturbations with GEARS. *Nature Biotechnology*, 42:927–935, 2024. doi: 10.1038/s41587-023-01905-6.
- [10] Haotian Cui, Chloe Wang, Hassaan Maan, et al. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21:1470–1480, 2024. doi: 10.1038/s41592-024-02201-0.
- [11] Christina V. Theodoris, Ling Xiao, Anant Chopra, et al. Transfer learning enables predictions in network biology. *Nature*, 618:616–624, 2023. doi: 10.1038/s41586-023-06139-9.
- [12] Minsheng Hao, Jing Gong, Xin Zeng, et al. Large-scale foundation model on single-cell transcriptomics. *Nature Methods*, 21:1481–1491, 2024. doi: 10.1038/s41592-024-02305-7.
- [13] Yanay Rosen, Yusuf Roohani, Ayush Agrawal, et al. Universal cell embeddings: A foundation model for cell biology. *bioRxiv*, 2023. doi: 10.1101/2023.11.28.568918. Preprint.

- [14] Hongzhi Wen, Wenzhuo Tang, Xinnan Dai, et al. Cellplm: Pre-training of cell language model beyond single cells. *bioRxiv*, 2023. doi: 10.1101/2023.10.03.560734. Preprint; also published at ICLR 2024.
- [15] Jérémie Kalfon, Jules Samaran, Gabriel Peyré, and Laura Cantini. scsprint: pre-training on 50 million cells allows robust gene network predictions. *Nature Communications*, 16:3607, 2025. doi: 10.1038/s41467-025-58699-1.
- [16] Xiaoning Qi, Lianhe Zhao, Chenyu Tian, et al. Predicting transcriptional responses to novel chemical perturbations using deep generative model for drug discovery. *Nature Communications*, 15:9256, 2024. doi: 10.1038/s41467-024-53457-1.
- [17] Ouyang Zhu and Jun Li. Scouter predicts transcriptional responses to genetic perturbations with large language model embeddings. *Nature Computational Science*, 6:21–28, 2025. doi: 10.1038/s43588-025-00912-8.
- [18] Stefan Peidli, Tessa D. Green, Ciyue Shen, et al. scPerturb: harmonized single-cell perturbation data. *Nature Methods*, 21:531–540, 2024. doi: 10.1038/s41592-023-02144-y.
- [19] Lukas Heumos, Yuge Ji, Lilly May, et al. Pertpy: an end-to-end framework for perturbation analysis. *Nature Methods*, 23:350–359, 2025. doi: 10.1038/s41592-025-02909-7.
- [20] Yan Wu, Esther Wershof, Sebastian M. Schmon, et al. Perturbench: Benchmarking machine learning models for cellular perturbation analysis. *arXiv preprint arXiv:2408.10609*, 2024. doi: 10.48550/arXiv.2408.10609. Preprint.
- [21] Pascal Notin, Aaron W. Kollasch, Daniel Ritter, et al. ProteinGym: large-scale benchmarks for protein fitness prediction and design. In *Advances in Neural Information Processing Systems*, volume 36, pages 64331–64379, 2023.
- [22] Daniel Esposito, Jochen Weile, Jay Shendure, et al. MaveDB: an open-source platform to distribute and interpret data from multiplexed assays of variant effect. *Genome Biology*, 20:223, 2019. doi: 10.1186/s13059-019-1845-6.
- [23] Oana Ursu, James T. Neal, Dhawal Bhatt, et al. Massively parallel phenotyping of coding variants in cancer with Perturb-seq. *Nature Biotechnology*, 40:896–905, 2022. doi: 10.1038/s41587-021-01160-7.
- [24] Hengshi Yu, Weizhou Qian, Yuxuan Song, and Joshua D. Welch. PerturbNet predicts single-cell responses to unseen chemical and genetic perturbations. *Molecular Systems Biology*, 21:960–982, 2025. doi: 10.1038/s44320-025-00131-3.
- [25] Sarah E. Cooper, Matthew A. Coelho, Magdalena E. Strauss, et al. scSNV-seq: high-throughput phenotyping of single nucleotide variants by coupled single-cell genotyping and transcriptomics. *Genome Biology*, 25:20, 2024. doi: 10.1186/s13059-024-03169-y.
- [26] Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 22: 1657–1661, 2025. doi: 10.1038/s41592-025-02772-6.
- [27] Rocío Viñas Torné, Maciej Wiatrak, Zoe Piran, et al. Systema: a framework for evaluating genetic perturbation response prediction beyond systematic variation. *Nature Biotechnology*, 44:1050–1059, 2026. doi: 10.1038/s41587-025-02777-8.
- [28] Zeming Lin, Halil Akin, Roshan Rao, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379:1123–1130, 2023. doi: 10.1126/science.ade2574.
- [29] Jun Cheng, Guido Novati, Joshua Pan, et al. Accurate proteome-wide missense variant effect prediction with alphamissense. *Science*, 381:eadg7492, 2023. doi: 10.1126/science.adg7492.

- [30] Sanjay R. Srivatsan, José L. McFaline-Figueroa, Vijay Ramani, et al. Massively multiplex chemical transcriptomics at single-cell resolution. *Science*, 367:45–51, 2020. doi: 10.1126/science.aax6234.
- [31] Britt Adamson, Thomas M. Norman, Marco Jost, et al. A multiplexed single-cell crispr screening platform enables systematic dissection of the unfolded protein response. *Cell*, 167:1867–1882.e21, 2016. doi: 10.1016/j.cell.2016.11.048.
- [32] Thomas M. Norman, Max A. Horlbeck, Joseph M. Replogle, et al. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science*, 365:786–793, 2019. doi: 10.1126/science.aax4438.
- [33] Joshua Meier, Roshan Rao, Robert Aber, et al. Language models enable zero-shot prediction of the effects of mutations on protein function. In *Advances in Neural Information Processing Systems*, volume 34, pages 29287–29303, 2021.
- [34] Omar Kana, Rance Nault, David Filipovic, Daniel Marri, Tim Zacharewski, and Sudin Bhattacharya. Generative modeling of single-cell gene expression for dose-dependent chemical perturbations. *Patterns*, 4(8):100817, 2023. doi: 10.1016/j.patter.2023.100817.
- [35] Zoe Piran, Niv Cohen, Yedid Hoshen, and Mor Nitzan. Disentanglement of single-cell data with biolord. *Nature Biotechnology*, 42:1678–1683, 2024. doi: 10.1038/s41587-023-02079-x.
- [36] Dominik Klein, Jonas Simon Fleck, Daniil Bobrovskiy, et al. CellFlow enables generative single-cell phenotype modeling with flow matching. *bioRxiv*, 2025. doi: 10.1101/2025.04.11.648220.
- [37] Abhinav Adduri, Dhruv Gautam, et al. Predicting cellular responses to perturbation across diverse contexts with state. *bioRxiv*, 2025. doi: 10.1101/2025.06.26.661135.